

Maximum-density kings on cylinders and planes (OEIS A137432, A061593, A061594)

The aspect-ratio transition on the cylinder;
all fixed orders on the square cylinder and the ratio conjecture;
pole bounds and inverses on the planar board

Three research reports of one external research session
(bundle Reports 174, 173 and 175), merged into one ProveIt report

All three Parts: 3 October 2026 merged: 6 October 2026

Abstract

A placement of hw kings on a $2h \times 2w$ board with no two attacking is a maximum-density placement: each 2×2 block holds exactly one king. This report prints three manuscripts on counting these placements. Part I treats the cylinder with $2h$ open rows and $2w$ cyclic columns, $A(h, w)$: a uniform marked-star approximation with relative error $\mathcal{O}(h^{-1})$ when w/h lies in a compact positive interval; a change of regime at $w/h = \log 2$, with the free energy, the subcritical equivalent, every fixed order above the transition with explicit first and second corrections, and a three-term critical-window expansion; three limit laws for the geometry of a uniform placement; Lambert- W inverse brackets along rational rays; and, for fixed height, squarefree denominators, the degree bound $\lfloor h/2 \rfloor + 1$ for every irreducible factor, attained at the heights $p - 3$. Part II treats the square cylinder $a_n = A(n, n)$, OEIS A137432: every fixed order $a_n = Cn^n(\sum_j c_j n^{-j})$ with $c_j \in \mathbb{Q}(e)$ (Part I's supercritical theorem at $\lambda = 1$), the exact c_3 , a boundary law with rate $\mathcal{O}(n^{-1})$, and the ratio expansion $a_{n+1}/(na_n) = e(1 + 1/(2n) + \dots)$, which proves the conjecture recorded in A137432 by Kotěšovec in 2011 that this ratio tends to e . Part III treats the ordinary $2h \times 2w$ board (A061593 at $h = 2$, A061594 at $h = 3$) with Part I's Gram machinery: every irreducible factor of the reduced fixed-height denominator has degree $d \leq \lfloor h/2 \rfloor + 1$ and exponent at most $h - 2d + 3$, so for even h every top-degree factor is simple; an exponential-polynomial formula with degree bounds; Wilf's dominant term $(\alpha_h w + \beta_h)(h + 1)^w$; and a Lambert- W threshold bracket with an exponentially small real error. Parts I and III prove two of the assertions recorded as conjectures on p. 83 of Kotěšovec's book (Conjectures 1 and 4) and the upper half of a third (Conjecture 2, on both boards), without first-proof claims. The write adds a proof of the two-sided bound $2(h + 1)^w \leq A(h, w) \leq 2^h(h + 1)^w$ with the cases of equality (the strict form of the upper bound for the square cylinder, as Part II quotes it from the book, fails exactly at $n = 1$), a proof that the dominant planar factor $1 - (h + 1)z$ has exponent exactly two at every height, and checks of the shared coefficients, the OEIS data and the inverses against the repository's transseries volume. The report is unrefereed, and nothing in it is formalized.

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Guide to this report

The three Parts are three manuscripts of the session bundle of Reports 1–243 (arrival commit 60f54ea06), placed in this directory by commit f7e9e5c2f (batch 109, cluster 109-KINGS). All three are dated 3 October 2026, and all three count maximum-density placements of kings: hw mutually nonattacking kings on a board of $2h \times 2w$ labelled squares, so that every one of the hw disjoint 2×2 blocks holds exactly one king.

Part	Bundle report	Delivered title	Labels
I	174 (base)	Aspect ratio transition for cylindrical kings: Uniform asymptotics and inverse thresholds for maximum density rectangles	mdk:asp:
II	173	Maximum density kings on a cylinder: All fixed order asymptotics and inverse thresholds for A137432	mdk:sq:
III	175	Planar Maximum Density Kings: Factor degree and pole multiplicity with fixed height asymptotics and inverses	mdk:pl:

One model. All three Parts encode a placement by a binary word $b \in \{0, 1\}^h$ (the column offsets of the kings, constant along each block row) and, in each block column, a threshold $c \in \{0, \dots, h\}$ (the row offsets are 0 above it and 1 below). Two successive thresholds are compatible according to a zero-one matrix M_b of size $h+1$, and $M_b = E_0 E_1$ factors through the incidence matrix C_b of the monochromatic components of the path coloured by b ; the bipartite graph of C_b is a caterpillar tree whose hubs are the runs of b . On the cylinder the count is a trace, $A(h, w) = \sum_b \text{tr}(M_b^w) = \sum_b \text{tr}((C_b C_b^T)^w)$ (Part I, (2.2) and (3.2); Part II, (12.4) and (13.1)). On the plane the transfer matrix is block upper triangular in the Boolean order of the words, with the same M_b as diagonal blocks, and the count is a boundary pairing $\mathbf{1}^T T^{w-1} \mathbf{1}$ (Part III, (5)). Parts I and II derive the asymptotics from one long run of b and the Schur-complement equation for the Perron eigenvalue of the caterpillar; Part III uses the smaller Gram matrix to bound the algebraic degree of the poles and the Boolean rank to bound their order.

Arrangement. The base of the merge is Report 174; its `Report174.tex` was staged as this `article.tex`, and it is printed first, as Part I, because the other two depend on it. Report 173’s main theorem is Part I’s supercritical theorem at aspect ratio $\lambda = 1$ (Theorem 1.2; the coefficients agree, see the note after Theorem 11.1), and its rectangular question (Section 21) is what Part I answers; Report 173 alone gives the exact c_3 , the ratio expansion that proves the OEIS conjecture, and its own uniform proof, so it is printed second, as Part II. Report 175 is printed third, as Part III: it reuses the component-incidence Gram factorization of Parts I and II (“reused from companion cylindrical analysis”, it says) on the planar board, and with Part I’s Appendix A it splits one family of Kotěšovec’s conjectures between the cylinder and the plane. The bundle numbers say the manuscripts were written in the order 173, 174, 175 (bundle index times 15:38, 16:16 and 16:55 UTC on 3 October 2026); Report 174 says that “No reading of an earlier square report is needed”, and none cites another in its bibliography.

Numbering. Sections are numbered continuously, and statements and equations within sections, as in all three manuscripts. Part I keeps Report 174’s Sections 1–10, its Appendices A–C and all its numbers. Report 173’s Section k is Section $k + 10$ here (Sections 11–21), and its statements and equations move with their sections (Report 173’s Corollary 1.3 is Corollary 11.3). Report 175’s Section k is Section $k + 21$ (Sections 22–30), and its statements move with their sections (its

Theorem 1.1 is Theorem 22.1); Report 175 numbers its equations consecutively through the manuscript, and Part III keeps those numbers, (1) to (31), so that an equation number without a dot belongs to Part III. Its one captioned table keeps its number, Table 1. The README maps all numbers. Section 31, Remark 20.1 and Remark 25.2 were added in the write; no delivered number of Part I or of Part III's equations moved, and no delivered statement number of Parts II and III moved within its section.

Notes added when the manuscripts were merged are set in small type and tagged [**write**], with their date (6 October 2026). They give cross-references between the Parts, record what the write checked, and record what later Parts and the write's own computations say about earlier text. No delivered statement was changed and no delivered symbol was renamed. Section 31 gathers the open questions of all three Parts under Vladimir's standing rule of 4 October 2026, with their sources, sketches and what is missing.

What is proved, and where

Statement	Where
<i>Exact encodings (prior: Knuth, Wilf, Matsuo, Brown; re-proved)</i>	
Cylinder: $A(h, w) = \sum_b \text{tr}(M_b^w)$, $M_b = E_0 E_1$, Gram and tree traces, caterpillar description	Part I, Sections 2–3; Part II, Sections 12–13
Plane: block upper triangular transfer, $P_h(w) = \mathbf{1}^\top T^{w-1} \mathbf{1}$, diagonal blocks $M_b = UV$ with VU a Gram matrix of dimension $\min(q+1, h-q+1)$	Part III, Lemmas 23.1, 24.1
<i>Cylinder, joint limit $w/h \rightarrow \lambda$ (Part I)</i>	
$A = S(h, w)(1 + \mathcal{O}(h^{-1}))$ uniformly for w/h in a compact positive interval	Theorem 1.1
Free energy $\Phi(\lambda)$, transition at $\lambda = \log 2$; subcritical equivalent; every fixed order above $\log 2$, $c_j \in \mathbb{Q}(\lambda, e^{-\lambda})$, explicit c_1, c_2	Theorem 1.2, Sections 5–6
Critical window $A/h^w = hG + \sqrt{h}H + J + \mathcal{O}(h^{-1/2})$, with the endpoint 29/12 and the spectral 4LG	Theorem 1.3, Section 7
Limit laws: finite, square-root (Rayleigh at $t = 0$) and linear defect; floor corrections	Theorems 8.1–8.3; Section 9
Integer first-crossing bracket along rational rays, higher-order inverse	Theorem 10.1, Section 10
<i>Cylinder, fixed height (Part I, Appendices)</i>	
Squarefree denominators (Kotěšovec’s Conjecture 1, p. 83); every irreducible factor of degree $\leq \lfloor h/2 \rfloor + 1$ (upper half of Conjecture 2, cylinder); equality at $h = p - 3$; $A(h, w) \sim 2(h+1)^w$ (prior, p. 192); small closed forms	Appendices A, B
$2(h+1)^w \leq A(h, w) \leq 2^h(h+1)^w$, equality cases; the book’s strict bound for a_n as Part II quotes it fails exactly at $n = 1$	[write] Remark 20.1
<i>Square cylinder, A137432 (Part II)</i>	
$a_n = Cn^n(\sum_{j \leq M} c_j n^{-j} + \mathcal{O}_M(n^{-M-1}))$, $C = 2e(e-1)^2/(e-2)^2$ (Kotěšovec’s constant), exact c_1, c_2, c_3	Theorem 11.1, (16.5)–(16.7); second route for c_1, c_2 and the all-order statement: Part I at $\lambda = 1$
Boundary law with total-variation rate $\mathcal{O}(n^{-1})$ and moment rates	Theorem 11.2; second route: Part I, Theorem 8.1 at $\lambda = 1$
$a_{n+1}/(na_n) = e(1 + 1/(2n) + (-c_1 - 1/24)/n^2 + \mathcal{O}(n^{-3}))$, hence the OEIS conjecture $a(n+1)/(na(n)) \rightarrow e$; integer bracket for the threshold	Corollary 11.3
<i>Plane, fixed height (Part III)</i>	
Factor degree $d \leq \lfloor h/2 \rfloor + 1$, exponent $\leq h - 2d + 3$, entrywise; top-degree factors simple for even h (Conjecture 4; upper half of Conjecture 2, planar)	Theorem 22.1, Corollary 22.2
Exponential-polynomial formula; Wilf’s dominant term with $\alpha_h \geq 1$	Theorems 26.1, 27.2
The factor $1 - (h+1)z$ has exponent exactly two at every height	[write] Remark 25.2
Lambert- W inverse, two-ceiling bracket, first exponentially small correction	Theorem 28.1, (30)
<i>Open</i>	
The regimes $w/h \rightarrow 0$; growing critical windows; effective constants and onsets (all inverses); subcritical corrections; the exact maximal degree at every height, cylinder and plane; low-degree multiplicities and cancellations in the plane; height-uniform planar asymptotics; large-order behaviour of c_j ; the history of the 2011 conjectures. See Section 31.	

The finite computations of the three packages (exact integer and rational arithmetic, all-width recurrence certificates) check identities, small counts and coefficient formulas; no asymptotic statement rests on them, and the floating-point residuals are diagnostics.

The A137432 conjecture and the leading constant

The write read the entry A137432 [2] on 6 October 2026 (internal format and revision history). Its name is “Number of ways to place n^2 nonattacking kings on a $2n \times 2n$ cylindrical chessboard.”, offset 0 with $a(0) = 1$ (prepended by Andrew Howroyd, 26 March 2023), and its author line reads “Vaclav Kotesovec, Aug 31 2011”. Two formula lines concern this report:

- “Conjecture: limit of $a(n+1)/(n \cdot a(n))$ as $n \rightarrow \infty$ is e .” The line is unsigned; the revision history shows it in revision #19, the entry’s first version under its present name, made by Vaclav Kotesovec on 31 August 2011. **Proved** by Part II, Corollary 11.3: the object of the conjecture is literally Part II’s a_n , and (11.4) gives $a_{n+1}/(na_n) = e(1 + 1/(2n) + \mathcal{O}(n^{-2}))$. Only the order $M = 0$ of Theorem 11.1 is needed for the limit: $a_{n+1}/(na_n) = (1 + 1/n)^{n+1}(1 + \mathcal{O}(n^{-1})) \rightarrow e$. Part I’s Theorem 1.2 at $\lambda_h = 1$ gives the same leading equivalent by its own proof. Part II says that its ratio expansion includes “the previously recorded ratio-limit conjecture”.
- “ $a(n) \sim c \cdot n^n$, where $c = 2 \cdot \exp(1) \cdot (\exp(1) - 1)^2 / (\exp(1) - 2)^2 = 31.1116835720490503682643922791052352237386275089\dots$ - Vaclav Kotesovec, Jul 29 2023, updated Mar 18 2024”. The entry gives no proof. Parts I and II prove it, with every fixed order, and both credit the constant to Kotěšovec (Part II, (11.1), prints 28 decimals, the entry 46; the write’s 70-digit evaluation agrees with all of them).

Nothing was submitted to the OEIS.

Kotěšovec’s book: the conjectures of p. 83 and the other cited pages

All three Parts cite V. Kotěšovec, *Non-attacking Chess Pieces*, sixth edition, 2013 [1], a large PDF on the author’s site. **The write did not read the book**, and neither did the intake: it is a large download that was not fetched without Vladimir’s permission. Everything below about its pages is therefore what the Parts state, with the Parts’ page numbers (the public file has been updated since 2013, as the bibliography of `a201513-sparse-chess-placements` notes, so page numbers may refer to the file the sources read on 3 October 2026). Part II’s source notes say that the relevant pages “were rendered and independently inspected” (`173-square-sources-REFERENCES.md`, item 4); Parts I and III name the pages they read.

The conjectures of p. 83, as the Parts describe them, and their status here.

Conjecture (Parts' description)	Status in this report
1 (dated 29 August 2011, per Part I): the fixed-height cylinder generating functions have squarefree minimal denominators; the book records Alekseyev's checks through $h = 20$	Proved , Part I, Appendix A ("We do not claim this is its first proof").
2 (dated 15 September 2011, per Part I): the maximal irreducible degree is $\lfloor h/2 \rfloor + 1$ "for the planar and horizontally cylindrical families"	Upper bound proved for the cylinder (Part I, (A.2)) and for the plane (Part III, Theorem 22.1); equality proved on the cylinder at the infinitely many heights $h = p - 3$, $p \geq 5$ prime (Part I). Equality at every height: open for both (Section 31, item 5).
3 (planar exponents, per Part III)	Not settled. Part III, Remark 25.1: the page prints the bound $\lfloor (h+1)/2 \rfloor$ while the same page lists exponent 2 at $h = 1$ and $h = 2$; the exponents are right (Part III's F_1 , F_2 ; the write's Remark 25.2); Part III does not "fix" the formula. Recorded as an inconsistency of the page as Part III reports it, not as a refutation (Section 31, item 6).
4 (planar, per Part III): for even h , factors of top degree are simple	Proved , Part III, Corollary 22.2; no first-proof claim.

So the family splits between the two boards: Conjecture 1 and the cylindrical half of Conjecture 2 are Part I's; the planar half of Conjecture 2, Conjecture 3 and Conjecture 4 concern Part III. Both Parts disclaim priority: the book credits a 2011 factor-generation method and an $h + 1$ degree bound to Alekseyev (p. 178, per Part I), whose full argument neither Part recovered, and the linked Zealint discussion [6] was inaccessible to them.

The other pages cited. Part I: pp. 82–83 and Section 2.6, pp. 169–196 (fixed-height cylinder families: exact formulas, factorizations, spectral power sums), p. 178 (Alekseyev), pp. 178–180 (simple-factor and equal-coefficient patterns), p. 192 (the fixed-height law $A(h, w) \sim 2(h+1)^w$, re-proved in Appendix A), p. 193 (leading and subleading roots), pp. 209–210. Part II: pp. 209–210, the cylinder of A137432, with "bounds, and limit questions". Part III: pp. 82–90 and 178.

The bounds of pp. 209–210. Part II, Section 20, reads there the lower bound $2(n+1)^n \leq a_n$ and a strict upper bound $a_n < 2^n(n+1)^n$, and notes that the strict bound "needs the restriction $n \geq 2$, since $a_1 = 4$ is equality". The arithmetic is right ($a_1 = 4 = 2^1 \cdot 2^1$; $a_1 = A(1, 1)$ by (B.1)). The write's Remark 20.1 proves both bounds for every rectangle, $2(h+1)^w \leq A(h, w) \leq 2^h(h+1)^w$, with equality in the upper bound exactly when $h = 1$ or $w = 1$; for the square this is the lower bound for all $n \geq 1$ and the strict upper bound for all $n \geq 2$. Whether the book states the restriction elsewhere could not be checked.

The OEIS entries and the data of the packages

A137432 (read 6 October 2026; revision history to #62 of 24 November 2025). Data $a(0), \dots, a(18)$; links: "Rintaro Matsuo, Table of n, a(n) for n = 0..384 (terms 1..31 from Alex V. Breger)" (the b-file), "Rintaro Matsuo, a(1)..a(800)" (a separate file, `a137432.txt`), Matsuo's repository with the polynomial-time algorithm, the Zealint blog (archived), and Kotěšovec's book, "pp. 209–210". Extensions: $a(11)$ – $a(12)$ by Kotěšovec, $a(13)$ – $a(31)$ by Alex V. Breger (September 2011). The ten counts printed in Part II, Section 19, and the diagonal of Part I's table in Appendix B agree with the entry's data.

The fixture of Part II. Report 173's `data/173-square-companion-fixtures-a137432_s`

`elected.json` holds $a_1, \dots, a_{10}$ and $a_{100}, a_{200}, a_{400}, a_{800}$; its `source_url` is the OEIS b-file <https://oeis.org/A137432/b137432.txt>, with a SHA-256 and the retrieval date 3 October 2026. The b-file covers $n = 0, \dots, 384$ (the entry’s link text; the intake confirmed the file itself on 6 October), so a_{400} and a_{800} cannot come from it. Part II’s text attributes the large terms to Matsuo’s table “through $n = 800$ ” [5], and its `173-square-sources-REFERENCES.md` (item 3) says the fixture records the URL and SHA-256 of that table. The fixture’s `source_url` therefore names the wrong file for the two largest terms, unless the b-file was longer on 3 October; which file the recorded SHA-256 belongs to was not determined (the write fetched neither file). The entry’s `a137432.txt`, “a(1)..a(800)”, is a second public home of such terms. The write’s only check of $a_{100}, \dots, a_{800}$ is indirect: the residuals $R_2(n), R_3(n)$ of Section 19 computed from the fixture (3000-digit arithmetic) reproduce every printed digit at $n = 800$ and move towards c_2 and c_3 as n doubles ($R_2 = 381.7, 455.3, 507.0, 538.9$ against $c_2 + c_3/n = 249.2, 412.8, 494.6, 535.4$ at $n = 100, 200, 400, 800$). This is consistent with the terms, not a verification of them.

Fixed-height cylinder entries (titles read 6 October 2026). A194644, “Number of ways to place $2n$ nonattacking kings on a $4 \times 2n$ cylindrical chessboard”, is the row $h = 2$ of Part I’s table: 12, 32, 90, 256, 732, 2102. A194647 (“ $5n \dots 10 \times 2n$ ”) is the row $h = 5$: 192, 708, 3036, 13932, 66532, 327192. A195656 (“ $12n \dots 24 \times 2n$ ”) is the row $h = 12$; its first term is $53248 = 13 \cdot 2^{12} = A(12, 1)$ by (B.1), and its twelfth term 222770819826574 is $a(12)$ of A137432, as it must be. These agree with Part I’s orientation (h open and fixed, w cyclic). (B.4) reproduces A194644’s first six terms.

Planar entries. A061593 (“Number of ways to place $2n$ nonattacking kings on a $4 \times 2n$ chessboard”, Antonio G. Astudillo, 22 May 2001) has the generating function $x(12 - 29x + 33x^2 - 9x^3)/((1 - 3x + x^2)(1 - 3x)^2)$, which is Part III’s F_2 (24) identically, and the closed form $(17n - 109)3^n + 2F(2n + 10)$, which gives Part III’s $(\alpha_2, \beta_2) = (17, -109)$. A061594 (“ $3n \dots 6 \times 2n$ ”, same author and date) has offset 0 with $a(0) = 1$; its generating function minus 1 is Part III’s F_3 (numerator in Section 27, denominator (15)), and its explicit formula has the term $(231n - 2377)4^n$, Part III’s (α_3, β_3) . (Checked by exact rational-function arithmetic.)

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The inverses and the transseries volume

The repository’s transseries volume [14] has a Part 0 apparatus for inverting rapidly growing functions. Its symbols are subscripted “vol” here where they meet this report’s letters ($a_{\text{vol}}, b_{\text{vol}}, \kappa_{\text{vol}}, d_{\text{vol}}, v_{\text{vol}}, X_{\text{vol}}, L_{\text{vol}}$). Statement by statement:

- Part II, the leading inverse $N = L/W_0(L)$, $N \log N = L$ (18.2), with $L = \log(x/C) > 0$: an *instance* of the factorial core `p0:prop:factorial-core` with $\kappa_{\text{vol}} = 1$, $d_{\text{vol}} = 0$, $L_{\text{vol}} = L$. The proposition gives $v_{\text{vol}} = W_0(L)$ and $X_{\text{vol}} = e^{v_{\text{vol}}} = L/W_0(L) = N$, the unique solution with $\log N > 0$.
- Part I, the leading inverse $N = (Y/p)/W_0(e^d Y/p)$ of $pN \log N + bN = Y$ (10.3), with $d = b/p$: an *instance* of the same proposition with $\kappa_{\text{vol}} = p$, $d_{\text{vol}} = b$ (Part I’s b ; Part I’s d is $d_{\text{vol}}/\kappa_{\text{vol}}$), $L_{\text{vol}} = Y > 0$. Then $v_{\text{vol}} = W_0(e^d Y/p)$ and $X_{\text{vol}} = e^{v_{\text{vol}} - d} = Y/(pv_{\text{vol}}) = N$, using $v_{\text{vol}} e^{v_{\text{vol}}} = e^d Y/p$; the side condition $p \log N + b > 0$ holds for large Y . Part II’s inverse is the ray $p = q = 1$ of this one ($b = 0, \beta = 0, D = C(1)$).

- Part III, the inverse $w_0(y)$ of $(\alpha x + \beta)n^x$ (25): an *instance*, after a change of variable, of the Lambert core `p0:thm:lambert-core`. With $X = x + \beta/\alpha$ and $b = \log n$ the equation $(\alpha x + \beta)n^x = y$ is $\log X + bX = \log(y/\alpha) + b\beta/\alpha$ for $X > 0$, the core with $a_{\text{vol}} = b = \log(h+1)$, $b_{\text{vol}} = 1 > 0$ and $L_{\text{vol}} = \log(y/\alpha) + b\beta/\alpha$; its branch rule gives the unique positive solution $X = b^{-1}W_0(be^{L_{\text{vol}}})$, which is (25) ($\alpha = \alpha_h \geq 1 > 0$ by Theorem 27.2).
- The one-step corrections: Part I's (10.5) (perturbation $\beta \log t$) and (10.8) (perturbation α_1/t), Part II's (18.3) (perturbation c_1/t) and Part III's (30) (perturbation R): each displayed correction is exactly the first-order term $X - \varrho(X)/\Phi'_0(X)$ of the perturbation-operator series `p0:eq:operator-series` (`p0:prop:operator-form`) about the corresponding core (for Part I, $\Phi'_0(N) = p(\log N + 1 + d)$; for Part II, $\Phi'_0(N) = 1 + \log N = D_N$). *Analogues*, not instances of a theorem of the volume: the proposition is a formal series, and the error bounds are the Parts' own. The same holds for Part II's fixed-order recursion for t_M (Section 18).
- Part III's threshold statements in Theorem 28.1: an *instance* of `p0:thm:staircase`. For n_1 larger than $\deg H_h$ and beyond the point where f_h increases, the real extension f_h restricted to $[n_1, \infty)$ is continuous, strictly increasing and equal to $P_h(w)$ at the integers, that is an admissible interpolation in the sense of `p0:def:three-inverses`, with inverse x_h ; Part III's identity $N_h(y) = \lceil x_h(y) \rceil$ is item (1) of the theorem (for y above $P_h(n_1 - 1)$, as P_h increases), (28) follows from it by monotonicity of the ceiling, and the agreement case of the bracket is item (2) with $x_* = w_0(y)$, $\eta = \varepsilon_h(y)$; (29) is item (1) with the interpolation g .
- Part I's Theorem 10.1 and Part II's bracket (11.5): proved directly at the integers from the comparison function g (resp. g_M) and the error K_0/n (resp. $D_M n^{-M-1}$), without an interpolation of the sequence; *not* instances of `p0:thm:staircase`. When the two ceilings agree they determine the threshold, the same conclusion as item (2) of that theorem, and the Parts' warnings that a single ceiling is not justified near a threshold describe its items (2) and (5).

None of the Parts claims novelty for the inversion mechanics; Part III calls its inversion “a standard inversion, not a priority claim”.

Notation across the three Parts

Common to all three Parts with the same meaning: h the number of block rows of a $2h \times 2w$ board (Part II has $h = w = n$), b the binary word, M_b the zero-one block matrix (Part III states its rule as “nondecreasing”, “nonincreasing”, Parts I and II as 0^*1^* , 1^*0^* : the same matrix), E_0, E_1 the component-membership matrices, Λ the squared Perron eigenvalue (Parts I and II), W_0 the positive real Lambert branch, tr , $\mathbb{Q}$. Identifications: Part I's $A(n, n)$ is Part II's a_n ; Part I's $C(1)$ is Part II's C and $c_j(1)$ is c_j (note after Theorem 11.1); Part I's resolvents $R_i(z)$ are Part II's $G_i(z)$; Part I's $\mathcal{H}_1$ is Part II's H ; Part I's α_j and S_1, T_1 are Part II's α_l and S, T ; Part III's X_0, X_1 are Part I's R_0, R_1 and Part II's R, S , and Part III's C is C_b ; Part III's G_b is the smaller of $C_b C_b^\top, C_b^\top C_b$. The symbols below change meaning between the Parts, or inside one; each Part fixes its meaning there, and each Part opens with a short table of its own readings. No symbol was renamed.

Symbol	Meanings
A, a	Part I: $A(h, w)$ the cylinder count; $\mathcal{A}_b$ the tree adjacency; $a_t(x)$ in (1.8); $a_n = A(qn, pn)$ along a ray (Section 10); a, b the bounds of w/h ; $a_h(\rho)$ spectral weights. Part II: $a_n = A(n, n)$; $A(T_b)$ the tree adjacency; $A(n) = a_n/(Cn^n)$; A_M, A_L, A_R . Part III: $a_{i,j}$ the row offset of a king (Parts I and II's x_{ij}); A the double-pole coefficient; $A_{\rho,j}$. <i>False reading</i> : Part I's a_n of Section 10 is not A137432 unless $p = q = 1$.
b, c	All: b the binary word. Part I: $b = p \log q + q\Phi(\lambda)$ in (10.1). Part III: $b_{i,j}$ the column offset (Parts I and II's y_{ij}); $(b, u), (c, v)$ states, so c is a second word; $b = \log n$ in Section 28. Parts I and II: c_j the threshold of block column j ((12.2)) and c_j the asymptotic coefficients. <i>False reading</i> : the thresholds c_j of Sections 2 and 12 are not the coefficients c_j .
h	Parts I and III: the height (open rows $2h$). Part II: $h = k + 1$ in Section 15; $h_a(q)$ sums and $h = q/(1 - q)$ in Section 17. Part I: $h_j(q)$. The transseries volume's letters are subscripted vol.
n	Part II: the board is $2n \times 2n$. Part I: the index along a ray. Part III: $n = h + 1$ (Lemma 27.1, Sections 27–28). <i>False reading</i> : Part III's n is not a board size.
w	Parts I and III: the width (block columns). Part II: w a pivot vertex in the proof of Lemma 13.1; $w_n(\omega), w_\infty$ masses. Part III: $w_0(y)$ the Lambert inverse.
q	Part I: $q = e^{-\lambda}$, and also the ray denominator in Section 10 (with $q_h = e^{-\lambda_h}$). Part II: $q = e^{-1}$. Part III: $q = b $, the Hamming weight (Lemma 24.1); q_i certificate coefficients.
d, D	Parts I and II: $d = 1 - k$. Part I: $D = h + d$ (Section 6), $D = h + 1 - k$ (Section 7), $D_q = qd/dq$, d_h the floor offset, $d = b/p$ and D the amplitude in Section 10. Part II: d also the number of outward runs (Section 16); $D = qd/dq$; $D_N = 1 + \log N$; D_M a constant. Part III: $d = [\mathbb{Q}(\rho) : \mathbb{Q}]$ the degree; $D = \text{diag}_b(M_b)$; D the certificate degree (Section 29).
e	Parts I and II: Euler's number ($E = e$ in (16.4)). Part III: e the exponent of a denominator factor, e_ρ a pole order. <i>False reading</i> : Part III's $e \leq h - 2d + 3$ is not about Euler's number.
k, K	Parts I and II: $k = h - m$ (resp. $n - m$) the defect. Part I: $K = h - M_h$ the random defect, K_∞ ; K a constant in Theorem 10.1. Part II: K the random boundary total, K_∞ , and $K = K_M(n)$ a cutoff. Part III: $k \in \{0, 1\}$ a colour, k a chain length, K the strictly upper part of T .
L	Part I: $L = \log 2$. Part II: $L = \log(x/C)$ in Section 18. Part III: L the recurrence onset (Section 29).
M, m	All: M_b . Parts I and II: m a run length, M a fixed order; Part I: M_h the longest run. Part III: m the index of s_m .
N	Part I: N_k the marked multiplicities; N the Lambert value. Part II: N the Lambert value; N_1, N_2, N_3 numerators. Part III: $N = (h + 1)2^h$ the number of states; $N_h(y)$ the threshold. <i>False reading</i> : Part III's N is a state count, not an inverse.
S, T	Part I: $S(h, w)$ the marked-star sum (not a count); $S_1 = B_1, T_1 = B_2$; T_b the tree; T a bound on $ t $. Part II: $S = s_L + s_R, T = B_2$; S the incidence matrix of the 1-components (with R); T_b, T_r trees. Part III: S_h the set of reciprocal poles; T the transfer matrix.
R	Part I: R_0, R_1 incidence matrices; $R_h(m)$ marked counts; $R_i(z)$ resolvents; $R_h = \sqrt{B \log h}$. Part II: R an incidence matrix; R_b the non-Perron remainder; $R_2(n), R_3(n)$ residuals; R a number of runs. Part III: $R(z) = (I - zD)^{-1}$, $R_b(z)$; $R_h(w)$ the subdominant remainder; $R(x) = f_h(x) - g(x)$.
C	Part I: $C(\lambda), C_b$, constants C, C_T . Part II: C the A137432 constant, C_b , constants. Part III: $C = X_0^\top X_1$ (Part I's C_b); C_h a constant.
F, G, H, J	Parts I and II: $F(q) = (1 - q)/(1 - 2q)$. Part III: $F_h(z)$ the generating function (Part I's fixed-height cylinder function is $\mathcal{F}_h$). Part I: G, H, J the critical coefficient functions; $\mathcal{H}_\lambda$. Part II: $G_i(z)$ resolvents; $H(u)$. Part III: G_b the Gram matrix; $H_h(z)$ the polynomial part and $H(z) = (1 - nz)^2 F_h(z)$; J the all-ones matrix.
B, W	Parts I and II: B_j branch moments; Part I also $B(\lambda), \mathcal{B}_h$ and a constant B . Parts I and II: $W, W_S, \dots$ weighted sums; W_0 Lambert.

Symbol	Meanings
α, β	Parts I and II: α_j (resp. α_l) the number of compositions. Part I: β a bound for the branch norms (Section 3) and $\beta \in \{3/2, 0\}$ in (10.1); $\alpha_1 = c_1(\lambda)/q$. Part III: α_h, β_h (also α, β) the dominant constants.
ρ, θ, η	Part I: $\rho < 1$ in (5.2); $\rho \in \mathcal{R}_h$ the spectral parameters; $\eta(y)$ the bracket width. Part II: $\rho = 2/e$; $\eta_M(x)$. Part III: $\rho \in S_h$ positive Gram eigenvalues; θ_h the subdominant base; η an exponential rate.
t, s, u, v	Part I: t_h the critical parameter, $t(y)$ the root; $s(\lambda)$; s_ℓ, t_ℓ run lengths; $u = c_j, v = c_{j+1}, u = 1/h, v_t$. Part II: $t_M(x)$; $t = m/n$; s_L, t_L, u_L the first three outward runs; $u = 1/n$. Part III: u, v thresholds; t a pivot vertex and $t = w_0(y)$; $s_m = \mathbf{1}^\top T^m \mathbf{1}$.
x, y	Parts I and II: x_{ij}, y_{ij} the row and column offsets; Part I: $x = k\varepsilon_h, y = \Lambda/h$, the target y . Part II: $y = \Lambda/n$; the target x . Part III: the target y ; x real.
r, ℓ	Part I: ℓ, r the boundary lengths; r_i run lengths; $\ell_j(\lambda)$. Part II: $l, r; \ell_j$; r a composition. Part III: $r = \theta_h/n$; $h = 2r$; r_b a dimension.
P, Q	Part I: $P_j(\lambda; \ell, r), P_6, P_8; Q_\ell, Q_r$ branch adjacencies. Part II: P_j . Part III: $P_h(w)$ the planar count; $\Pi = J/n$; $Q(z)$ a denominator. <i>False reading</i> : Part III's P_h is a count, not a coefficient polynomial.
λ, Φ, g, f	Part I: λ the aspect ratio, $\lambda_h = w/h$; $\Phi(\lambda)$ the free energy; $g(t)$; f_t, f_λ ; g_0, g_1, g_2 . Part II: $\lambda = \sqrt{\Lambda}$ in the proof of Lemma 14.1; g_M ; $f(t)$ a rate. Part III: $g(x) = (\alpha x + \beta)n^x$; f_h the real extension. The volume's core is written Φ_0 (front matter only).
I, U, V, X	Part I: $I_j(t)$ integrals; (U_t, V_t) limit variables; X_t . Part III: I the identity; U, V factors with $UV = M_b$; X_k incidence matrices.

Provenance and merge decisions

Three manuscripts of the session bundle of Reports 1–243, all dated 3 October 2026. The author lines of Reports 173 and 174 are empty; Report 175's reads “Research report 175”, which is also its PDF author field. None names an author or pins a repository commit. Report 175's code provenance says it was “prepared 2026-10-03 with OpenAI assistance” (175-planar-README_CODE.md, data/175-planar-data-certificates.json, code/175-planar-regenerate.py); Reports 173 and 174 name no tool. The intake found no “prepared for private review” wording in any text file, PDF metadata or PDF text. They arrived in commit 60f54ea06 and were placed by f7e9e5c2f (batch 109, cluster 109-KINGS).

- **Part I**, bundle Report 174, archive `Rectangular_Kings_Aspect_Ratio_Transition_Source.zip` (477,514 bytes; 13 files in the wrapper `Report174/`; `Report174.tex`, 805 lines, 22 pages); the base. Contributes the joint-limit theory, the critical window, the limit laws, the ray inverses and the fixed-height structure of the cylinder.
- **Part II**, bundle Report 173, archive `Cylindrical_Kings_All_Order_Asymptotics_and_Inverses_Source.zip` (523,435 bytes; 23 files; `Report173.tex`, 627 lines, 17 pages). Contributes a second proof of the square case, the exact c_3 , the ratio expansion and the A137432 conjecture, and the large-term residual diagnostics.
- **Part III**, bundle Report 175, archive `Planar_Kings_Pole_Bounds_and_Inverses_Source.zip` (546,797 bytes; 21 files; `report.tex`, 489 lines, 13 pages). Contributes the planar pole theorem and its fixed-height consequences, with all-width generating-function certificates through $h = 6$.

Where the merge had to choose:

1. *Base and order.* Report 174 first, as the base; 173 second; 175 third (see the Guide). Two separate reports (cylinder 173+174; plane 175 alone) were considered at intake and not taken, because Part III reuses Part I's Gram machinery and the two split one family of conjectures.
2. *Repeated arguments.* Parts I and II prove the same encoding, factorization, caterpillar description, Schur equation, uniform analytic disk and boundary law; Part III proves the factorization again for the plane. The texts are independent paraphrases with different detail (Part II's disk has explicit radii; Part I carries λ ; Part III adds the Boolean-rank count). Each Part's proof is complete only with its own version, so each is printed as delivered, with **[write]** pointers; nothing was dropped.
3. *Front matter, numbering, running heads and bibliography.* The three title blocks and abstracts are printed at the heads of their Parts; one table of contents replaces three; the delivered running heads ("Report 17x" and the title) are dropped, as are Report 174's redefined title block and the delivered link colours; Report 175's list spacing (`\setlist`) is applied from Part III on. Part I's Appendices A–C stay where Report 174 put them, after its Section 10, and numbering returns to arabic for Part II. The three bibliographies are merged into one, keeping every detail of every entry; where the Parts cite the same work with different details (the book's pages, Wilf's DOI or journal link, Brown's arXiv or journal version, the two A137432 descriptions) all are kept and attributed. The keys OEIS (Parts I and II) became `oeisA137432`, Part I's `RectOEIS` became `oeisRect`, and Part III's `a061593`, `a061594` became `oeisA061593`, `oeisA061594`; the case variants `Kotesovec/kotesovec`, `Wilf/wilf`, `Brown/brown` were unified. The write added the transseries volume.
4. *Write additions.* This front matter, the reading-conventions tables of the Parts, Remarks 20.1 and 25.2, Section 31, the dated notes, section labels for Part III, and the label prefixes. Everything else is delivered text.

What was checked, and what was not

The report is unrefereed and not formalized. At intake (dossier of batch 109, 6 October 2026) the three manuscripts were read in full and no false mathematical step was found; the delivered suites reproduced their recorded outputs on copies (Report 173's default, `-0`, extended and residual runs and the optional SymPy derivation of c_3 ; Report 174's default, `-0`, extended, symbolic and packaging tests; Report 175's default, `-0`, extended, guard and SymPy regeneration through $h = 4$); an independent row-mask count gave $a_1, \dots, a_4$, the corner $A(h, w)$, $h, w \leq 4$, of Part I's table (including $A(1, 2) = 8 \neq A(2, 1) = 12$), and P_1, P_2, P_3 for $w \leq 8$; $J(0)$, $H(0)$, $G(0)$, $A(h, 2)$ and $A(2, w)$ were checked by hand.

The write (6 October 2026): proved Remarks 20.1 and 25.2; checked by exact rational-function arithmetic (SymPy 1.14.0) that Part I's $c_1(\lambda)$ and $c_2(\lambda)$ of (6.11), (6.12) at $\lambda = 1$, $q = e^{-1}$ are Part II's (16.5), (16.6), that Part I's operator forms (6.9), (6.10) equal its explicit forms, and that Part II's shorter form (16.8) equals Part I's $c_2(1)$; evaluated C , c_1 , c_2 , c_3 , $G(0)$, $H(0)$, $J(0)$ to 45 digits (every printed digit agrees); recomputed $R_2(n)$, $R_3(n)$ from the fixture; checked F_2 and F_3 against A061593 and A061594 and (α_2, β_2) , (α_3, β_3) by Part III's extraction formula; checked the two-sided bounds against $a_1, \dots, a_{10}$; read the OEIS entries named above; compared the staged files with a fresh extraction of the three archives (47 files, no difference); and reran, on fresh extractions, Report 173's `run_checks.py` and `residuals.py --precision 80` and Report 174's

`run_checks.py`, and Report 175's `verify.py` on a copy of this directory: every output reproduced the shipped file byte for byte after removing carriage returns.

What was *not* done: the analytic proofs (uniform approximation, localization, the analytic disks, Euler–Maclaurin with Gaussian bounds, the limit laws, the Boolean-chain pole count, Wilf's double pole) were read and their displayed algebra spot-checked, but they are the manuscripts' proofs, not an independent verification. Kotěšovec's book, Matsuo's repository and tables, Knuth's notes, Wilf, Larsen, Brown and Bagno et al. were not read by the write; the fixture's large terms and its SHA-256 were not compared with any source file. The PDF and ZIP build tools were not run.

Relation to neighbouring reports and formal projects

The only other report of the collection on nonattacking kings is **a201513-sparse-chess-placements** (repository search, 6 October 2026). It treats the sparse regime: K_n counts n kings on an ordinary $n \times n$ board (A201513), density tending to zero, against the carrier $B_n = e^{-9/2} n^{2n} / n!$, by a cluster expansion for finite move sets. This report treats the dense regime, hw kings on $2h \times 2w$ boards (density $1/4$), with carriers Cn^n or $(h+1)^w$, by a block encoding and transfer spectra. No lemma is shared; that report's K_n and B_n are not this report's a_n , $A(h, w)$, K or B_j . Both cite Kotěšovec's book, at different pages. The transseries volume [14] is related to the inverses as stated above. Placement in the collection confers no formal status: no Lean or Rocq file in the repository treats king placements, and no statement of this report is formalized.

Part I

Aspect ratio transition for cylindrical kings: uniform asymptotics and inverse thresholds for maximum density rectangles

Bundle Report 174, archive `Rectangular_Kings_Aspect_Ratio_Transition_Source.zip`, delivered as `Report174/Report174.tex` (22 pages), the base of the merge. Delivered title block “Aspect ratio transition for cylindrical kings / Uniform asymptotics and inverse thresholds for maximum density rectangles / Report 174 / 3 October 2026” (no author line). Printed as delivered, with `[write]` notes; section, appendix, statement and equation numbers are those of the delivery.

Abstract of this Part (as delivered). Let $A(h, w)$ count placements of hw nonattacking kings on a labeled $2h \times 2w$ board with columns identified cyclically and rows left open. As $h, w \rightarrow \infty$ with $w/h \rightarrow \lambda > 0$, the dominant binary run changes from a linear defect to a finite defect at $\lambda = \log 2$. A uniform marked-star approximation, with relative error $\mathcal{O}(h^{-1})$, proves the free energy and the subcritical equivalent. Above the transition we prove an expansion to every fixed algebraic order and give explicit first and second corrections. In the critical window $w = (\log 2)h + t\sqrt{h}$, with the actual lattice-compatible t retained, we obtain $A(h, w)/h^w = hG(t) + \sqrt{h}H(t) + J(t) + \mathcal{O}(h^{-1/2})$, including the endpoint and spectral contributions to J . We also prove three geometric limit laws, Lambert W inverse brackets along rational rays, and exact fixed-height spectral consequences. Existing rectangular formulas and fixed-height asymptotics of Kotěšovec, earlier factor-generation work credited to Alekseyev, Wilf’s transfer coordinates, Matsuo’s cylindrical encoding, and Kotěšovec’s square leading constant are explicitly distinguished from the joint-limit analysis. This is an unrefereed report with reproducible checks and no exhaustive priority claim.

Reading conventions of Part I ([write]; see the front matter for all Parts)

$A(h, w)$	$2h$ open rows, $2w$ cyclic columns; h is the open direction (fixed in Appendix A), w the wrapped one; $A(h, w) \neq A(w, h)$ in general. Part II’s a_n is $A(n, n)$.
L, λ_h, q, t_h	$L = \log 2$ (not Part II’s L); $\lambda_h = w/h$; $q = e^{-\lambda}$, except in Section 10, where q is the ray denominator; $t_h = (w - Lh)/\sqrt{h}$.
c_j	In Section 2 the threshold of block column j ; elsewhere the asymptotic coefficients $c_j(\lambda)$.
$S(h, w), \mathcal{F}_h$	The marked-star sum, an approximation and not a count; $\mathcal{F}_h(z)$ the fixed-height cylinder generating function (Part III’s F_h is planar).
d, D	$d = 1 - k$ and $D = h + d$ (Section 6); $D = h + 1 - k$ (Section 7); d_h a floor offset; in Section 10, $d = b/p$ and D an amplitude.

1 Model and the phase diagram

The squares have coordinates (r, s) with $0 \leq r < 2h$ and $s \in \mathbb{Z}/(2w\mathbb{Z})$. Two distinct kings attack when their row difference and circular column distance are both at most one. Rows do not wrap. We count placements on this labeled board, without dividing by rotations, reflections, or translations. Throughout h, w are positive integers. A partition into hw disjoint 2×2 blocks gives an upper bound of hw kings; occupying every other row and column attains it. Thus $A(h, w)$ is exactly a maximum-density count.

The square diagonal $A(n, n)$ is [OEIS A137432](#). Fixed-height rectangular families already appear in Kotěšovec's book, notably Section 2.6, pp. 169–196 [1], and in OEIS entries such as A194644, A194647, and A195656 [3]. The orientation convention is important: h always refers here to the open direction and w to the wrapped direction. In general $A(h, w) \neq A(w, h)$.

Set

$$L = \log 2, \quad \lambda_h = \frac{w}{h},$$

and define the explicitly evaluable marked-star sum

$$S(h, w) = 2(h+1)^w + \sum_{m=1}^{h-1} (h-m+3)2^{h-m-1}(m+1)^w. \quad (1.1)$$

The terminology records a marked run in an exact binary encoding; S itself is not an exact count of king placements.

Theorem 1.1 (Uniform marked-star approximation). *For every fixed $0 < a \leq b < \infty$, uniformly over integers w with $a \leq w/h \leq b$,*

$$A(h, w) = S(h, w)(1 + \mathcal{O}_{a,b}(h^{-1})). \quad (1.2)$$

No assumption that the longest run exceeds $h/2$ is required.

Theorem 1.2 (Free energy and the two fixed regimes). *If $\lambda_h \rightarrow \lambda > 0$, then*

$$\lim_{h \rightarrow \infty} \frac{1}{h} \log \frac{A(h, w)}{h^w} = \Phi(\lambda), \quad \Phi(\lambda) = \begin{cases} L - \lambda + \lambda \log(\lambda/L), & 0 < \lambda < L, \\ 0, & \lambda \geq L. \end{cases} \quad (1.3)$$

The maximizing run fraction is $s(\lambda) = \min(\lambda/L, 1)$. The function Φ is continuously differentiable; its second derivative has left limit $1/L$ and right limit zero at L .

Uniformly on each compact subset of $0 < \lambda_h < L$,

$$A(h, w) = h^{w+3/2} B(\lambda_h) e^{h\Phi(\lambda_h)} (1 + \mathcal{O}(h^{-1})), \quad B(\lambda) = \frac{\sqrt{2\pi\lambda}}{L} \left(1 - \frac{\lambda}{L}\right). \quad (1.4)$$

Uniformly on each compact subset of $\lambda_h > L$, for every fixed integer $M \geq 0$,

$$A(h, w) = h^w C(\lambda_h) \left(\sum_{j=0}^M \frac{c_j(\lambda_h)}{h^j} + \mathcal{O}_M(h^{-M-1}) \right), \quad (1.5)$$

where

$$F(q) = \frac{1-q}{1-2q}, \quad C(\lambda) = 2e^\lambda F(e^{-\lambda})^2, \quad c_0 = 1, \quad c_j(\lambda) \in \mathbb{Q}(\lambda, e^{-\lambda}). \quad (1.6)$$

The notation for the coefficient field means that each c_j is a rational function of the displayed variables, evaluated on the supercritical interval.

Section 6.3 gives c_1, c_2 explicitly and a finite algorithm for every fixed order. The quantifier “fixed M ” does not assert convergence of the formal infinite series, uniformity for growing M , or a complete expansion in exponentially small sectors.

Theorem 1.3 (Critical crossover through constant order). *Let*

$$t_h = \frac{w - Lh}{\sqrt{h}}, \quad I_j(t) = \int_0^\infty x^j e^{-tx - Lx^2/2} dx.$$

For every fixed $T < \infty$, uniformly for $|t_h| \leq T$,

$$\frac{A(h, w)}{h^w} = hG(t_h) + \sqrt{h}H(t_h) + J(t_h) + \mathcal{O}_T(h^{-1/2}). \quad (1.7)$$

Writing $f_t(x) = e^{-tx - Lx^2/2}$, set

$$a_t(x) = t + Lx - \frac{t}{2}x^2 - \frac{L}{3}x^3, \quad (1.8)$$

$$b_t(x) = L(-\frac{1}{2} + x^2 - \frac{1}{4}x^4) + t(x - \frac{1}{3}x^3). \quad (1.9)$$

Then

$$G(t) = I_1(t), \quad (1.10)$$

$$H(t) = \int_0^\infty [3 + xa_t(x)]f_t(x) dx = 4I_0(t) - I_2(t) - \frac{t}{6}I_3(t), \quad (1.11)$$

$$J(t) = \frac{29}{12} + \int_0^\infty \left\{ 3a_t(x) + x \left[b_t(x) + \frac{a_t(x)^2}{2} \right] \right\} f_t(x) dx + 4LG(t). \quad (1.12)$$

The 29/12 is an endpoint correction. The $4LG(t)$ is the first nontrivial spectral correction, which is absent from the marked-star sum.

All refined formulas use the exact parameters λ_h or t_h , not a rounded nominal parameter. Section 9 gives the explicit floor corrections. In particular,

$$A(h, \lfloor Lh \rfloor) \sim \frac{h^{\lfloor Lh \rfloor + 1}}{L}.$$

What is carried over and what is analyzed here. Binary/threshold coordinates already occur in Wilf's planar block decomposition [8] and are presented explicitly for the cylinder by Matsuo [4]. We prove the cylindrical encoding here to make the report self-contained. Kotěšovec already gives exact rectangular fixed-height spectral formulas and the law $A(h, w) \sim 2(h+1)^w$ for fixed h and $w \rightarrow \infty$ [1, p. 192]. His p. 178 credits a 2011 direct factor-generation method and a degree bound to Alekseyev. The linked full argument was inaccessible in the bounded source review, so priority for the spectral reduction is not asserted. The square constant

$$C(1) = \frac{2e(e-1)^2}{(e-2)^2}$$

is explicitly credited to Kotěšovec in A137432, dated 29 July 2023 and updated 18 March 2024 [2]. The present focus is the joint aspect-ratio limit, uniform approximation, crossover, geometry, and inverse consequences. No reading of an earlier square report is needed.

[write] **6 October 2026.** The earlier square report is Report 173, printed here as Part II. On the diagonal $h = w = n$ one has $\lambda_h = 1 > L$, so Theorem 1.2 applies with $\lambda_h = 1$ and is Part II's Theorem 11.1: $C(1)$ is Part II's C (11.1), and $c_1(1)$, $c_2(1)$ of (6.11), (6.12) are Part II's (16.5), (16.6) as elements of $\mathbb{Q}(e)$ (for c_1 multiply numerator and denominator of (6.11) by e^4 at $q = e^{-1}$: they become $N_1(e)$ and $2(e-1)^2(e-2)^2$;

for c_2 the write checked the identity by exact rational-function arithmetic). Since an asymptotic expansion in powers of n^{-1} is unique, the two theorems together give $c_j(1) = c_j$ for every j ; Part II's c_3 (16.7) is therefore $c_3(1)$, which this Part does not compute. Further second routes between the two Parts: Theorem 8.1 at $\lambda = 1$ is Part II's Theorem 11.2; Theorem 10.1 on the ray $p = q = 1$ (where $\beta = 0$, $b = \Phi(1) = 0$, $D = C(1)$ and $g(t) = t \log t$) is Part II's bracket (11.5) with $M = 0$, and Section 10's higher-order inverse on that ray is Part II's g_M .

2 The exact geometric encoding

Index the 2×2 blocks by (i, j) , with $0 \leq i < h$ and j modulo w . The unique king in block (i, j) has position

$$(2i + x_{ij}, 2j + y_{ij}), \quad x_{ij}, y_{ij} \in \{0, 1\}.$$

If $y_{ij} = 1$ and $y_{i,j+1} = 0$, the kings in these neighboring blocks attack: their column distance is one and their row distance is at most one. Hence the cyclic sequence y_{ij} cannot descend, so it is constant in j :

$$y_{ij} = b_i, \quad b = (b_0, \dots, b_{h-1}) \in \{0, 1\}^h. \quad (2.1)$$

For $w = 1$ constancy is automatic and involves no comparison of a king with itself. Similarly $x_{ij} = 1, x_{i+1,j} = 0$ would cause a vertical attack. Thus, for each j , the row offsets are nondecreasing in i , and there is a unique threshold $c_j \in \{0, \dots, h\}$ with

$$x_{ij} = 0 \quad (i < c_j), \quad x_{ij} = 1 \quad (i \geq c_j).$$

For two successive block-columns with thresholds $u = c_j$ and $v = c_{j+1}$, a down-right diagonal attack occurs exactly when

$$u \leq i < v - 1, \quad (b_i, b_{i+1}) = (1, 0).$$

If $u < v$, excluding these attacks says that $b_u, \dots, b_{v-1}$ is nondecreasing. The opposite diagonal direction similarly says, when $u > v$, that $b_v, \dots, b_{u-1}$ is nonincreasing. If $u = v$, there is no extra condition. These conditions exclude every possible attack; nonneighboring blocks cannot contain attacking squares. When $w = 2$, the two directions between block-columns meet the same pair, and both ordered conditions are imposed by the cyclic product below. When $w = 1$, its single diagonal condition is automatic and the vertical rule already excludes the relevant attacks.

Define an $(h+1) \times (h+1)$ zero-one matrix M_b by these allowed transitions. The list of thresholds is cyclic and labeled. Consequently

$$A(h, w) = \sum_{b \in \{0,1\}^h} \text{tr}(M_b^w), \quad w \geq 1. \quad (2.2)$$

There is no factor w and no division by w . The identity is not an empty-width convention: at $w = 0$ its right side is $2^h(h+1)$.

3 From a transfer matrix to a caterpillar

Take the path with vertices $0, \dots, h$ and edge colors $b_0, \dots, b_{h-1}$. Let $E_c[u, v]$ be one if the entire interval from u to v has color c , including the empty interval. Then

$$M_b = E_0 E_1. \quad (3.1)$$

Indeed an intermediate vertex must be in the same color-0 component as u and the same color-1 component as v . There is at most one such vertex, since two would require a nonempty interval of both colors. For $u < v$ it exists precisely when the interval is 0^*1^* ; for $u > v$, the increasing-order interval must be 1^*0^* . An intermediate vertex outside the interval would again require a nonempty segment of both colors. This proves the factorization entry by entry.

Let R_0, R_1 be the indicator matrices of the color-0 and color-1 path components. We have $E_c = R_c R_c^\top$. Put

$$C_b = R_0^\top R_1.$$

Distinct components of different colors intersect in at most one path vertex, so C_b is a zero-one matrix. Its bipartite incidence graph T_b is connected: successive original path vertices give edges sharing the component of their connecting color. It has $h + 1$ edges, one for each path vertex. If there are n_0 zero edges and n_1 one edges, its number of vertices is $(n_1 + 1) + (n_0 + 1) = h + 2$. Thus it is a tree.

By cyclicity of trace (or the equality of nonzero spectra of rectangular products), for every positive w ,

$$\text{tr}(M_b^w) = \text{tr}((C_b C_b^\top)^w) = \frac{1}{2} \text{tr}(\mathcal{A}_b^{2w}), \quad \mathcal{A}_b = \begin{pmatrix} 0 & C_b \\ C_b^\top & 0 \end{pmatrix}. \quad (3.2)$$

In particular the eigenvalues contributing to the trace are nonnegative squared adjacency eigenvalues. The zero eigenvalues are irrelevant at positive powers.

If b has successive run lengths $r_1, \dots, r_s$, T_b is a caterpillar. For $s \geq 2$ its spine consists of s vertices, with pendant-leaf counts

$$r_1, r_2 - 1, \dots, r_{s-1} - 1, r_s.$$

For $s = 1$ it is the star with $h + 1$ leaves. This follows by contracting the alternating monochromatic components along the original path; each internal run loses one endpoint to each neighboring spine edge. In every case a run of length m gives a hub of degree $m + 1$. Deleting that hub removes $m + 1$ of the tree's $h + 1$ edges; hence its remaining branches contain $h - m$ edges in total.

3.1 Bounds and the rooted Schur equation

Write $\Lambda_b = \|\mathcal{A}_b\|^2$. If the longest run has length m , separate the caterpillar adjacency into pendant-star edges and spine-path edges. These have norms at most $\sqrt{m + 1}$ and 2, respectively. Therefore

$$\Lambda_b \leq (\sqrt{m + 1} + 2)^2. \quad (3.3)$$

Now mark a run of length m and delete its hub. Apart from isolated pendant leaves, there are at most two rooted branches, with adjacency matrices Q_ℓ, Q_r . Let β bound their squared norms and let

$$B_j = \langle e_\ell, Q_\ell^{2j} e_\ell \rangle + \langle e_r, Q_r^{2j} e_r \rangle, \quad j \geq 1,$$

with absent branches contributing zero. The spectral theorem gives

$$0 \leq B_j \leq B_1 \beta^{j-1}. \quad (3.4)$$

If $\Lambda = \Lambda_b > \beta$, the resolvent expansion in the Schur complement at $\sqrt{\Lambda}$ gives

$$\Lambda = m + 1 + \sum_{j \geq 1} B_j \Lambda^{-j}, \quad 0 \leq \Lambda - (m + 1) \leq \frac{B_1}{\Lambda - \beta}. \quad (3.5)$$

To see the constant term, each neighbor of the deleted hub contributes $1/\sqrt{\Lambda}$ to the first resolvent term; there are $m+1$ such neighbors. Odd rooted moments vanish because each branch is bipartite. Multiplication by $\sqrt{\Lambda}$ yields the displayed equation. The lower bound $\Lambda \geq m+1$ follows from the hub star.

If $k = h - m$, then also $\beta \leq k$: the squared adjacency norm of any bipartite graph is the squared operator norm of its incidence matrix, at most its squared Frobenius norm, namely its number of edges. If the first two outward run lengths on each branch are s_ℓ, t_ℓ and s_r, t_r , missing lengths being zero, rooted walk counts give

$$B_1 = s_\ell + s_r, \quad B_2 = s_\ell^2 + t_\ell + s_r^2 + t_r. \quad (3.6)$$

Interlacing after deleting the hub bounds every adjacency eigenvalue other than the Perron pair by $\sqrt{\beta}$ in absolute value. Thus

$$0 \leq \text{tr}(M_b^w) - \Lambda_b^w \leq (h+1)\beta^w. \quad (3.7)$$

These estimates keep the non-Perron trace contribution separate from the Perron expansion.

4 Marked runs and the uniform approximation

Mark a particular run of length m in a word of length h . Set $k = h - m$ and let ℓ, r be the total lengths on its left and right, so $\ell + r = k$. Each boundary is encoded by its outward run composition. The inner bit is forced opposite to the central bit. Let

$$\alpha_0 = 1, \quad \alpha_j = 2^{j-1} \ (j \geq 1), \quad \sum_{j \geq 0} \alpha_j q^j = F(q) = \frac{1-q}{1-2q}.$$

There are two choices of the central bit. Therefore the exact number of marked words with this run length is

$$R_h(h) = 2, \quad R_h(h-k) = 2 \sum_{\ell=0}^k \alpha_\ell \alpha_{k-\ell} = (k+3)2^{k-1} \quad (1 \leq k < h). \quad (4.1)$$

This counts marked runs of exact length m , not just longest runs. A word with several eligible runs appears several times. The distinction will be dealt with before a one-hub approximation is used.

Lemma 4.1 (Bounded average branch moment). *For the marked words of defect k , with $B_1 = s_\ell + s_r$,*

$$\mathbb{E}_k B_1 = \frac{4k}{k+3} \quad (k \geq 1), \quad \mathbb{E}_0 B_1 = 0. \quad (4.2)$$

Proof. The weighted series for the first run length on one outward composition is $F(q) \sum_{s \geq 1} s q^s = F(q)q/(1-q)^2$. For the two boundaries, before the central-bit factor, the total series is

$$2F(q)^2 \frac{q}{(1-q)^2} = \frac{2q}{(1-2q)^2}.$$

Its coefficient at q^k is $k2^k$. The count coefficient is $[q^k]F(q)^2 = (k+3)2^{k-2}$. The central-bit factor cancels in their ratio. $\square$

Proof of Theorem 1.1. Fix $a \leq w/h \leq b$. All constants below may depend on a, b . Choose a small fixed $\delta > 0$ with $L + a \log \delta < 0$, allowing strict slack in this inequality.

Small longest runs. For words whose longest run satisfies $m < \delta h$, the norm bound (3.3), the 2^h choices of word, and the at most $h + 1$ nonzero trace terms give

$$\sum_{\max \text{ run} < \delta h} \text{tr}(M_b^w) \leq h^w e^{-ch}$$

for sufficiently large h . Strict slack absorbs the $\mathcal{O}(\sqrt{h})$ norm correction and all polynomial factors. The portion of the marked-star sum with $m < \delta h$ has the same estimate: there are at most $h2^h$ marked runs and their weights are $(m + 1)^w$. The baseline $S(h, w) \geq 2(h + 1)^w \geq 2h^w$ makes these errors exponentially small relative to S .

Two macroscopic runs. Choose $0 < \epsilon < \delta/16$. For a marked run of length $m \geq \delta h$, states having a second run of length at least ϵh are at most

$$\text{poly}(h) 2^{h-m-\epsilon h}$$

in number. Indeed specify the two disjoint run intervals and their bits, then allow all other positions freely. There are polynomially many choices of interval endpoints. This upper bound compared with $R_h(m) \geq 2^{h-m-1}$ is exponentially small. If there is no space for the second run the count is zero.

For the actual king contribution, mark a longest run m . Uniformly for $m \geq \delta h$, (3.3) yields

$$\frac{\Lambda_b^w}{(m + 1)^w} \leq e^{C\sqrt{h}}.$$

The trace dimension and possible choices of longest run contribute only polynomial factors. The entropy penalty $2^{-\epsilon h}$ dominates these factors and $e^{C\sqrt{h}}$. Summing over m bounds all such bad words by $Se^{-c\epsilon h}$; the same argument removes bad marked states from S . This explicitly includes ties and multiple large hubs.

One macroscopic run. Every remaining word has a unique run $m \geq \delta h$, with all other runs shorter than ϵh . It corresponds to exactly one good marked state. On deleting its hub, (3.3) applied to the remaining caterpillar branches gives

$$\beta \leq (\sqrt{\epsilon h + 1} + 2)^2 \leq 2\epsilon h$$

for large h . Since $\Lambda \geq m + 1 \geq \delta h$, the Schur bound gives

$$0 \leq \Lambda - (m + 1) \leq C \frac{B_1}{h}.$$

Moreover $B_1 \leq 2\epsilon h$. Hence

$$0 \leq \frac{\Lambda^w}{(m + 1)^w} - 1 \leq C \frac{B_1}{h}. \quad (4.3)$$

For precision, the logarithm of the ratio is bounded by CB_1/h , a uniformly bounded number, and $e^x - 1 \leq C'x$ on such a bounded interval. The non-Perron bound (3.7) is exponentially small relative to $(m + 1)^w$ because $\beta/(m + 1) \leq 2\epsilon/\delta < 1$ and $w \geq ah$.

For each m , the sum of B_1 over good marked states is at most its sum over all marked states, which is less than $4R_h(m)$ by Lemma 4.1. Summing (4.3) therefore produces an error $\mathcal{O}(S/h)$. Each good word contributes at least its star weight, so the same decomposition gives both inequalities in (1.2). There is no remaining multiplicity or trace-dimension loss. $\square$

5 Evaluation below the transition

The exponential rate of a marked-run summand with $s = m/h$ is

$$f_\lambda(s) = (1 - s)L + \lambda \log s, \quad 0 < s \leq 1.$$

Its maximum is attained at $s = \lambda/L$ if $\lambda < L$, and at $s = 1$ otherwise. Polynomial multiplicities do not change the logarithm divided by h . Finite-sum upper bounds and a term near the maximizing integer give (1.3), using Theorem 1.1. The same observation also explains why the transition occurs at L .

For a sharp subcritical equivalent, substitute $j = m + 1$ in (1.1):

$$S(h, w) = 2(h + 1)^w + 2^h \sum_{j=2}^h (h - j + 4) j^w e^{-Lj}. \quad (5.1)$$

When w/h lies in a fixed compact subinterval of $(0, L)$, the saddle $j = w/L$ is a linear distance below h . Extending separately the positive sums of $j^w e^{-Lj}$ and $j^{w+1} e^{-Lj}$ to infinity changes each by an exponentially small relative error. Their resulting difference remains of order h times the first sum, since $1 - w/(Lh)$ stays bounded away from zero. Thus cancellation does not invalidate this operation. The $j = 1$ and star-endpoint terms are also exponentially negligible.

For any integer $p \geq 1$, the exact pole series is

$$\sum_{j \geq 1} j^p e^{-Lj} = p! \sum_{\ell \in \mathbb{Z}} (L + 2\pi i \ell)^{-p-1} = \frac{\Gamma(p+1)}{L^{p+1}} (1 + \mathcal{O}(\rho^p)) \quad (5.2)$$

for a fixed $\rho < 1$. One derivation differentiates the partial-fraction expansion of $(e^z - 1)^{-1}$ p times at $z = L$; after differentiation the pole series is absolutely convergent, including $p = 1$. Alternatively it follows by Poisson summation applied to $x^p e^{-Lx}$ on the positive half-line. The nonzero poles are uniformly farther from L than the pole at zero; summing their absolute values gives the displayed geometric estimate.

Applying (5.2) with $p = w, w + 1$ yields

$$A(h, w) = \frac{2^h \Gamma(w+1)}{L^{w+1}} \left(h + 4 - \frac{w+1}{L} \right) (1 + \mathcal{O}(h^{-1})). \quad (5.3)$$

For S the error here is exponentially small; for A the relative $\mathcal{O}(h^{-1})$ comes from the spectral approximation. Uniform Stirling expansion gives (1.4). The constants 4 and $1/L$ inside the parentheses do not by themselves give the full first correction to A : spectral structure also contributes at that order.

6 All fixed orders above the transition

Assume $a \leq \lambda_h \leq b$ with $a > L$. We give the uniform argument, not just a formal expansion at each fixed boundary.

6.1 Localization and summable tails

For every fixed small $\epsilon > 0$, words whose longest run is at most $(1 - \epsilon)h$ contribute $\mathcal{O}(h^w e^{-c\epsilon h})$. For m/h bounded away from zero this follows from (3.3), the bound $2h2^{h-m}$ on words with a marked longest run, and

$$(1 - m/h)L + \lambda_h \log(m/h) < 0.$$

The inequality is uniform away from $m/h = 1$ because $\lambda_h \geq a > L$. For sufficiently small m/h , counting all 2^h words gives a strictly negative rate as in Section 4.

For $m = h - k$ with $k \leq \epsilon h < h/2$, the long run is unique. Put $d = 1 - k$, $D = h + d$. By (3.5) and $B_1 \leq k$, $\beta \leq k$,

$$0 \leq \Lambda - D \leq \frac{k}{h - 2k + 1}. \quad (6.1)$$

The inequality $\log x \leq x - 1$ consequently gives

$$(\Lambda/h)^w \leq C e^{-ak} \quad (0 \leq k \leq \epsilon h). \quad (6.2)$$

The constant is uniform in $h, k, \lambda_h \in [a, b]$. Multiplicity is $\mathcal{O}((k+1)2^k)$, so this is summable with ratio at most $2e^{-a} < 1$. Separately, by (3.7), the normalized non-Perron sum is bounded by a polynomial factor times

$$\exp\{h[\epsilon L + a \log \epsilon]\},$$

which is exponentially small for sufficiently small ϵ . After that separate bound, no trace dimension factor is carried into the Perron coefficient expansion.

For any desired fixed algebraic order, truncate at $k \leq A_M \log h$, where A_M is sufficiently large. Equation (6.2) makes the omitted tail smaller than the requested power of h^{-1} .

6.2 Uniform analytic remainder

Put $u = 1/h$ and $y = \Lambda/h$. The Schur equation becomes

$$y = 1 + du + \sum_{j \geq 1} B_j u^{j+1} y^{-j}. \quad (6.3)$$

The bound $B_j \leq k^j$ shows, by geometric-series estimates, that on a complex disk $|u| \leq c/(k+1)$ the right side is a contraction in $|y - 1| \leq 1/4$, for a fixed sufficiently small $c > 0$. It maps the disk into itself and has derivative of absolute value less than one there. Its fixed point is analytic and agrees with the positive Perron solution on small positive u . In particular

$$y = 1 + du + \mathcal{O}(ku^2), \quad \frac{\log y}{u} - d = \mathcal{O}((k+1)^2 |u|),$$

where the apparent singularity at $u = 0$ is removable; for $k = 0$ the first remainder is exactly zero.

On the smaller disk $|u| \leq c'/(k+1)^2$, the analytic function

$$\mathcal{H}_\lambda(u) = \exp \left\{ \lambda \left(\frac{\log y}{u} - d \right) \right\}$$

is uniformly bounded for $\lambda \in [a, b]$. Cauchy estimates imply that its order- j coefficient is $\mathcal{O}_j((k+1)^{2j})$, and that

$$\mathcal{H}_\lambda(1/h) = \sum_{j=0}^M P_j(\lambda; \ell, r) h^{-j} + \mathcal{O}_M((k+1)^{2M+2} h^{-M-1}) \quad (6.4)$$

provided $1/h \leq c'/(2(k+1)^2)$. This holds uniformly for $k \leq A_M \log h$ once h is large. The coefficient notation includes the full outward compositions, not just their lengths. Multiplying by $e^{\lambda d}$ and summing the remainder against the marked multiplicities costs only a fixed moment of the geometric tail $(2e^{-a})^k$. Thus the total error is $\mathcal{O}_M(h^{-M-1})$ without a logarithmic loss. Coefficient tails can also be extended to all finite boundary compositions. This proves (1.5) with the leading sum

$$2e^\lambda \sum_{\ell, r \geq 0} \alpha_\ell \alpha_r e^{-\lambda(\ell+r)} = 2e^\lambda F(e^{-\lambda})^2.$$

6.3 A finite coefficient procedure

For a rooted branch with outward run lengths $s_1, \dots, s_v$, define formal series recursively by

$$R_{v+1}(z) = 1, \quad R_i(z) = \frac{1}{1 - z(s_i - 1 + R_{i+1}(z))}. \quad (6.5)$$

Eliminating pendant leaves in the rooted caterpillar proves this diagonal-resolvent identity. Padding missing run lengths by zero preserves $R = 1$. The coefficient of z^j involves only the first j runs, with the right value when a branch terminates earlier. The branch sum satisfies

$$B_j = [z^j](R_{\ell,1}(z) + R_{r,1}(z) - 2).$$

Formal substitution in (6.3), followed by logarithm and exponentiation, computes each P_j as a finite rational polynomial in k , λ , and finitely many outward run lengths.

Write $D_q = q d/dq$ and

$$h_j(q) = \sum_{s \geq 1} s^j q^s \quad (j \geq 0), \quad h_0 = \frac{q}{1-q}, \quad h_1 = \frac{q}{(1-q)^2}, \quad h_2 = \frac{q(1+q)}{(1-q)^3}.$$

For a monomial in the first v run lengths with last exponent $a_v > 0$, its weighted sum over finite compositions is

$$F(q) \prod_{i=1}^v h_{a_i}(q).$$

The constant monomial has sum $F(q)$. Products combine independent left and right boundaries; powers of their total size k are implemented by D_q . Since $h_{j+1} = D_q h_j$, all these sums are rational functions of q . This proves the claimed rationality at every fixed order, and specifies a finite coefficient algorithm rather than relying on interpolation.

For the first two corrections let $S_1 = B_1$ and $T_1 = B_2$. Expansion gives

$$\Lambda = h + d + S_1 h^{-1} + (T_1 - dS_1) h^{-2} + \mathcal{O}_k(h^{-3}), \quad (6.6)$$

$$\begin{aligned} \frac{\Lambda^w}{h^w} = e^{\lambda_h d} & \left[1 + \frac{\lambda_h(S_1 - d^2/2)}{h} \right. \\ & \left. + \frac{\lambda_h(T_1 - 2dS_1 + d^3/3) + (\lambda_h^2/2)(S_1 - d^2/2)^2}{h^2} + \mathcal{O}_k(h^{-3}) \right]. \end{aligned} \quad (6.7)$$

The fixed- k notation in this display is made uniform upon summation by (6.4).

For a compact operator form put $\Delta = 1 - D_q$ and

$$\begin{aligned} W &= F^2, & W_S &= 2F^2 h_1, \\ W_{SS} &= 2F^2 h_2 + 2F^2 h_1^2, & W_T &= 2F^2(h_2 + h_0 h_1). \end{aligned} \quad (6.8)$$

These are the weighted sums of $1, S_1, S_1^2, T_1$ for two boundaries. Holding λ fixed under differentiation and evaluating at $q = e^{-\lambda}$,

$$c_1(\lambda) = \frac{\lambda}{W} \left(W_S - \frac{1}{2} \Delta^2 W \right), \quad (6.9)$$

$$c_2(\lambda) = \frac{1}{W} \left\{ \lambda \left(W_T - 2\Delta W_S + \frac{1}{3} \Delta^3 W \right) + \frac{\lambda^2}{2} \left(W_{SS} - \Delta^2 W_S + \frac{1}{4} \Delta^4 W \right) \right\}. \quad (6.10)$$

In explicit rational form,

$$c_1(\lambda) = -\frac{\lambda(4q^4 - 40q^3 + 45q^2 - 12q + 1)}{2(1-q)^2(1-2q)^2}, \quad q = e^{-\lambda}, \quad (6.11)$$

and

$$c_2(\lambda) = \frac{\lambda P_6(q)}{3(1-q)^3(1-2q)^3} + \frac{\lambda^2 P_8(q)}{8(1-q)^4(1-2q)^4}, \quad (6.12)$$

where

$$\begin{aligned} P_6(q) &= 8q^6 - 380q^5 + 690q^4 - 401q^3 + 69q^2 - 5q + 1, \\ P_8(q) &= 16q^8 - 976q^7 + 2920q^6 - 2940q^5 + 977q^4 + 14q^3 + 2q^2 - 6q + 1. \end{aligned}$$

At $\lambda = 1$ these specialize to the square corrections

$$c_1(1) = -13.572052159046386\dots, \quad c_2(1) = 576.331755295050712\dots$$

The large size of these coefficients warns against assuming that a few terms give accurate finite-height numerics without a remainder estimate.

[write] 6 October 2026. These are Part II's (16.5) and (16.6), with the same digits (note at the end of Section 1); the write also checked that the operator forms (6.9), (6.10) equal (6.11), (6.12) identically in λ and q . Part II adds $c_3(1)$.

7 The critical expansion

Fix $|t| \leq T$ and write $w = Lh + t\sqrt{h}$, using only pairs for which w is integral. For intermediate algebra the real exponent is also meaningful for the star sum, but it is not a king count unless w is integral. In this section put

$$\varepsilon_h = h^{-1/2}, \quad x = k\varepsilon_h, \quad D = h + 1 - k.$$

7.1 Localization and spectral perturbation

Choose a small fixed $\epsilon > 0$, for example $1/4$. Words with no run longer than $(1-\epsilon)h$ are exponentially negligible after division by h^w . For $m/h = r$ away from zero their upper rate is

$$(1-r)L + (w/h) \log r = L(1-r + \log r) + o(1),$$

strictly negative on each compact subinterval bounded away from $r = 1$. For $m \leq h/4$, counting all words and using (3.3) gives the negative upper rate $L + L \log(1/4)$. This covers the remaining endpoint. The $o(1)$ is uniform for bounded t .

Near the long run, $k \leq \epsilon h$, non-Perron eigenvalues give an exponentially small aggregate contribution by (3.7): its rate is at most $\epsilon L + (w/h) \log \epsilon < 0$. We may thus focus on the Perron eigenvalue of the unique long run. Here the stronger moment bound is

$$0 \leq B_j \leq B_1 k^{j-1}.$$

Writing $S_1 = B_1 \leq k$, the Schur equation gives

$$\Lambda - D = \frac{S_1}{h} + \mathcal{O}\left(\frac{S_1(k+1)}{h^2}\right). \quad (7.1)$$

Indeed isolate B_1/Λ , bound the remaining sum by $S_1 k/[\Lambda(\Lambda - k)]$, and compare $1/\Lambda$ to $1/h$ using (6.1). It follows that

$$(\Lambda/D)^w = 1 + \frac{w}{h} \frac{S_1}{h} + \mathcal{O}\left(\frac{S_1(k+1)}{h^2}\right). \quad (7.2)$$

The exponent is uniformly bounded; its squared-error contribution is absorbed using $S_1^2 \leq k S_1$. A uniform second-moment assumption is unnecessary.

Let $N_0 = 2$ and $N_k = (k+3)2^{k-1}$ for $k \geq 1$, and define the truncated base sum

$$\mathcal{B}_h = 2(1 + 1/h)^w + \sum_{1 \leq k \leq \epsilon h} N_k (D/h)^w. \quad (7.3)$$

The choice of rounding at the upper limit is exponentially immaterial. Averaging (7.2) with (4.2) gives

$$\frac{A(h, w)}{h^w} = \mathcal{B}_h + \frac{w/h}{h} \sum_{1 \leq k \leq \epsilon h} N_k (D/h)^w \frac{4k}{k+3} + \mathcal{O}_T(h^{-1/2}). \quad (7.4)$$

For completeness, the error is bounded by

$$\frac{C}{h^2} \sum_{k \geq 1} (k+1)^2 \exp\left(\frac{Tk}{\sqrt{h}} - c \frac{k^2}{h}\right) = \mathcal{O}_T(h^{-1/2}),$$

using the Gaussian estimate established next. The correction sum in (7.4) equals

$$4LG(t) + \mathcal{O}_T(h^{-1/2}). \quad (7.5)$$

The leading Riemann sum has this value because the exact average $4k/(k+3)$ cancels the multiplicity factor $k+3$. This is the full spectral contribution at constant order.

7.2 A summably uniform Taylor expansion

The inequality $\log(1-u) \leq -u - u^2/2$, with $u = (k-1)/h$, yields

$$2^k (D/h)^w \leq C_T e^{Tx - cx^2}, \quad 1 \leq k \leq \epsilon h. \quad (7.6)$$

The replacement of D by Λ adds only a bounded factor by (7.1). Therefore the region $x > R_h = \sqrt{B \log h}$ has a tail smaller than any specified inverse power of h if the fixed B is sufficiently large. All needed polynomial moments are dominated uniformly for $|t| \leq T$.

For $x \leq R_h$, Taylor's theorem gives

$$kL + w \log(1 + (1-k)/h) = L - tx - \frac{L}{2} x^2 + \varepsilon_h a_t(x) + \varepsilon_h^2 b_t(x) + \mathcal{O}_T(\varepsilon_h^3 (1+x^5)). \quad (7.7)$$

The a_t, b_t are exactly (1.8)–(1.9). Since $\varepsilon_h(1 + R_h^3) \rightarrow 0$, exponentiation yields

$$2^k(D/h)^w = 2f_t(x) \left[1 + \varepsilon_h a_t(x) + \varepsilon_h^2 \left(b_t(x) + \frac{1}{2} a_t(x)^2 \right) + \mathcal{O}_T(\varepsilon_h^3(1 + x^9)) \right]. \quad (7.8)$$

The integrated error after multiplication by $k + 3$ and summation is $\mathcal{O}_T(\varepsilon_h)$ without logarithmic loss: use the displayed polynomial times the Gaussian majorant, whose moments are uniformly finite, and choose the discarded tails stronger than the desired remainder.

To make the pointwise bookkeeping explicit, put $v_t = b_t + a_t^2/2$ and

$$g_0 = x f_t, \quad g_1 = (3 + x a_t) f_t, \quad g_2 = (3 a_t + x v_t) f_t.$$

For $k \geq 1$ the summand is

$$N_k(D/h)^w = \varepsilon_h^{-1} g_0(x) + g_1(x) + \varepsilon_h g_2(x) + 3\varepsilon_h^2 v_t(x) f_t(x) + \mathcal{O}_T(\varepsilon_h^3(k + 3)(1 + x^9) f_t(x)). \quad (7.9)$$

The explicit fourth term has total $\mathcal{O}_T(\varepsilon_h)$ after summation. It cannot be hidden inside the smaller displayed pointwise remainder; it is absorbed only after summing.

7.3 Endpoint terms and the constant coefficient

Euler–Maclaurin with uniform Gaussian derivative bounds gives, for the functions needed here,

$$\sum_{k \geq 1} g(k\varepsilon_h) = \varepsilon_h^{-1} \int_0^\infty g(x) dx - \frac{g(0)}{2} - \frac{\varepsilon_h g'(0)}{12} + \mathcal{O}_T(\varepsilon_h^3), \quad (7.10)$$

or its lower-order version where sufficient. Here

$$g_0(0) = 0, \quad g'_0(0) = 1, \quad g_1(0) = 3.$$

The exceptional $k = 0$ term is $2(1 + 1/h)^w = 4 + \mathcal{O}_T(\varepsilon_h)$. Applying (7.10) to (7.9) therefore gives

$$\mathcal{B}_h = hG(t) + \sqrt{h}H(t) + \left(4 - \frac{1}{12} - \frac{3}{2} + \int_0^\infty g_2(x) dx \right) + \mathcal{O}_T(h^{-1/2}).$$

The endpoint combination is $4 - 1/12 - 3/2 = 29/12$. Adding (7.5) proves Theorem 1.3. Extending the formula $(k + 3)2^{k-1}$ to $k = 0$ would give a wrong constant.

7.4 Evaluation and matching of the coefficient functions

Completing the square gives

$$I_0(t) = \sqrt{\frac{\pi}{2L}} e^{t^2/(2L)} \operatorname{erfc}\left(\frac{t}{\sqrt{2L}}\right), \quad G(t) = \frac{1 - tI_0(t)}{L}. \quad (7.11)$$

Integration by parts gives

$$LI_1 + tI_0 = 1, \quad LI_{j+1} + tI_j = jI_{j-1} \quad (j \geq 1). \quad (7.12)$$

Thus G, H, J are explicit polynomial combinations of $t, L, I_0(t)$. The recurrence also reduces the integral expression for H to (1.11). At the center,

$$\begin{aligned} G(0) &= \frac{1}{L} = 1.4426950408889634\dots, \\ H(0) &= \left(4 - \frac{1}{L}\right) \sqrt{\frac{\pi}{2L}} = 3.8497251900678098\dots, \\ J(0) &= \frac{119}{12} - \frac{8}{3L} + \frac{2}{3L^2} = 7.457059211633169\dots \end{aligned} \tag{7.13}$$

These are coefficient values, not an assertion that an integer pair has $w = Lh$.

For $t \rightarrow +\infty$, rescaling $x = y/t$ in G and expanding the Gaussian under its Laplace integral gives

$$G(t) = t^{-2} - 3Lt^{-4} + 15L^2t^{-6} + \mathcal{O}(t^{-8}).$$

For $t \rightarrow -\infty$, completing the square and localizing at $x = -t/L$ gives

$$G(t) \sim \frac{-t}{L} \sqrt{\frac{2\pi}{L}} e^{t^2/(2L)}.$$

The first matches $C(L + \delta) \sim \delta^{-2}$; the second matches the formal nearcritical limit of (1.4). These are matching statements about coefficient functions and neighboring fixed-regime formulas. They do not extend the proven compact- t remainder to an arbitrary growing window.

8 Three limiting geometries

Choose a maximum-density placement uniformly. Its associated word has weight $\text{tr}(M_b^w)$. Write M_h for the longest run and $K = h - M_h$ for the defect. On the high-probability event of a unique macroscopic run, record its central bit and its two outward compositions. Let ℓ, r be their lengths, so $\ell + r = K$. Set both boundary lengths and their fractions to zero on the exceptional event, and set $\ell/K = 0$ when $K = 0$; these conventions have no effect on the critical or subcritical limits.

The proof of Theorem 1.1 gives more than a scalar estimate. On the good marked-state set, the sum of absolute differences between the true state weights and the star weights is $\mathcal{O}(S/h)$; the omitted mass is exponentially small. After normalization, the two probability measures differ in total variation by $\mathcal{O}(h^{-1})$. One may adjoin a single exceptional state to make the comparison a statement on a common space. The central bit is exactly fair and independent of the run-composition data, by word complementation, which preserves the tree spectrum.

Theorem 8.1 (Supercritical finite defect). *If $\lambda_h \rightarrow \lambda > L$, set $q = e^{-\lambda}$. The left and right outward run compositions converge jointly to independent finite compositions, a specified composition of total length j having probability $q^j/F(q)$. Their central bit is independent and fair. The limiting defect satisfies*

$$\mathbb{E}[z^{K_\infty}] = \frac{F(qz)^2}{F(q)^2}, \quad |z| \leq 1. \tag{8.1}$$

Uniformly on compact supercritical intervals, comparison to the law with the exact $q_h = e^{-\lambda_h}$ has total variation error $\mathcal{O}(h^{-1})$, and every fixed nonnegative integer defect moment has error $\mathcal{O}(h^{-1})$.

Proof. The leading state weight is $2e^{\lambda_h} q_h^{\ell+r}$ before separating the two equally likely bits. Section 6 bounds the first-order error by a summable polynomial in K times $(2e^{-a})^K$. The same remains

true after multiplication by any fixed power of K . Normalizing gives the stated total variation and moment bounds, and summing the two independent size series gives (8.1). $\square$

Equivalently, on each side the number of runs is geometric on $\{0, 1, \dots\}$ with ratio $q/(1-q)$; conditional on that number, run lengths are independent with probabilities $(1-q)q^{s-1}$, $s \geq 1$. Literal boundary bits are constrained by the central bit; independence concerns outward compositions or boundaries encoded relative to it.

Theorem 8.2 (Critical square-root defect). *If $t_h \rightarrow t \in \mathbb{R}$, then $K/\sqrt{h}$ converges in distribution, and in every fixed nonnegative integer moment, to X_t with density*

$$p_t(x) = \frac{xe^{-tx-Lx^2/2}}{G(t)}, \quad x > 0. \quad (8.2)$$

Jointly the boundary lengths satisfy

$$(\ell/\sqrt{h}, r/\sqrt{h}) \implies (U_t, V_t), \quad p(u, v) = \frac{e^{-t(u+v)-L(u+v)^2/2}}{G(t)}, \quad u, v > 0. \quad (8.3)$$

Thus ℓ/K converges to a uniform variable on $[0, 1]$, independent of $K/\sqrt{h}$ and the fair central bit. At $t = 0$, X_0 is Rayleigh with density $Lxe^{-Lx^2/2}$.

Proof. The leading marked weight is $(k+3)f_t(k/\sqrt{h})$ and its total is asymptotic to $hG(t)$. Riemann sums give (8.2). The Gaussian bounds from Section 7, also valid with a fixed polynomial multiplier and a bounded spectral factor, give the asserted moment convergence for the true weights. For a given $k \geq 2$, each interior split $1 \leq \ell \leq k-1$ has multiplicity 2^{k-1} ; each endpoint split has twice this multiplicity. The endpoints have total mass $\mathcal{O}(h^{-1/2})$. The interior two-dimensional Riemann sum gives (8.3). Transforming (u, v) to $(x = u + v, z = u/(u + v))$ introduces Jacobian x and factorizes the density as $p_t(x) dz$. $\square$

Theorem 8.3 (Subcritical linear defect and Gaussian run). *If $\lambda_h \rightarrow \lambda \in (0, L)$, then*

$$\frac{M_h}{h} \rightarrow \frac{\lambda}{L} \quad \text{in probability}, \quad \frac{M_h - w/L}{\sqrt{h}} \implies N\left(0, \frac{\lambda}{L^2}\right). \quad (8.4)$$

The fraction $\ell/(h - M_h)$ converges to a uniform variable on $[0, 1]$, independent of the Gaussian limit and central bit. All other runs have maximum $\mathcal{O}(\log h)$ with probability tending to one.

Proof. The saddle for $j = m + 1$ in (5.1) is $j = w/L$, and the smooth prefactor $h - j + 4$ changes its location by only $\mathcal{O}(1)$. Taylor expansion of $w \log j - Lj$ at the saddle has quadratic term $-L^2(j - w/L)^2/(2w)$. Standard Gaussian localization follows directly from the strict concavity of this exponent on a fixed proportional neighborhood of the saddle, with exponentially small complementary tails. This proves (8.4); replacing j by $m = j - 1$ affects the scaled center by $o(1)$. The uniform-split calculation is the same as above, with k now proportional to h , so endpoint mass tends to zero. Conditional on a boundary length, all its compositions are equally likely; their cuts are independent fair choices among the internal positions. A union bound gives probability at most $Ch2^{-v}$ for any noncentral run to have length at least v . Taking v a sufficiently large multiple of $\log h$ proves the last assertion for the marked measure, and total variation transfer proves it for placements. $\square$

The transition is between linear, square-root, and finite defect scales. A unique macroscopic run persists for every fixed positive aspect ratio; it does not disappear in the subcritical regime.

9 Floors and exact parameter conventions

For a fixed $0 < \lambda < L$, write $w = \lfloor \lambda h \rfloor = \lambda h + d_h$ with $-1 < d_h \leq 0$. Since $\Phi'(\lambda) = \log(\lambda/L)$,

$$\frac{A(h, w)}{h^w} = h^{3/2} B(\lambda) e^{h\Phi(\lambda)} \left(\frac{\lambda}{L} \right)^{d_h} (1 + \mathcal{O}(h^{-1})). \quad (9.1)$$

Discarding the factor $(\lambda/L)^{d_h}$ is not generally an asymptotic equivalent.

For fixed $\lambda > L$, the cleanest all-order statement is to keep λ_h in (1.5). At the first two orders, Taylor expansion with $w = \lambda h + d_h$ gives

$$\begin{aligned} \frac{A(h, w)}{h^w} = C(\lambda) & \left[1 + \frac{c_1 + d_h C'/C}{h} \right. \\ & \left. + \frac{c_2 + d_h c'_1 + d_h (C'/C) c_1 + (d_h^2/2) C''/C}{h^2} + \mathcal{O}(h^{-3}) \right], \end{aligned} \quad (9.2)$$

where every unmarked function and derivative on the right is evaluated at λ . Uniformity holds for bounded d_h , including floor sequences. In particular replacing $C(\lambda_h)$ by $C(\lambda)$ preserves the leading equivalent, but not all higher coefficients.

For a critical floor sequence $w = \lfloor Lh + t\sqrt{h} \rfloor$, put $d_h = w - Lh - t\sqrt{h}$. The actual parameter is $t_h = t + d_h/\sqrt{h}$. Theorem 1.3 and Taylor expansion yield the floor-aware formula

$$\frac{A(h, w)}{h^w} = hG(t) + \sqrt{h} [H(t) + d_h G'(t)] + J(t) + d_h H'(t) + \frac{d_h^2}{2} G''(t) + \mathcal{O}_T(h^{-1/2}). \quad (9.3)$$

All derivatives exist and are bounded on compact intervals by their Gaussian-integral representations. Even the coefficient of $\sqrt{h}$ oscillates with the floor. For $t = 0$, the center values in (7.13) must therefore be combined with these derivative terms before making a next-order assertion about $\lfloor Lh \rfloor$.

10 Lambert W inversion along rational rays

Fix positive integers p, q and set

$$\lambda = p/q, \quad a_n = A(qn, pn), \quad n \geq 1.$$

A rational p/q cannot equal $\log 2$, so the ray is in one of the two fixed regimes. Define

$$\begin{aligned} \beta &= \begin{cases} 3/2, & \lambda < L, \\ 0, & \lambda > L, \end{cases} \quad b = p \log q + q\Phi(\lambda), \\ D &= \begin{cases} q^{3/2} B(\lambda), & \lambda < L, \\ C(\lambda), & \lambda > L. \end{cases} \end{aligned} \quad (10.1)$$

The forward theorems give

$$\log(a_n/D) = pn \log n + bn + \beta \log n + \mathcal{O}(n^{-1}). \quad (10.2)$$

This proves eventual strict increase: the main consecutive difference is $p \log n + \mathcal{O}(1)$ and the difference of two individually $\mathcal{O}(1/n)$ errors is still $\mathcal{O}(1/n)$.

For a sufficiently large target y , put $Y = \log(y/D)$ and $d = b/p$. Let W_0 be the positive real Lambert W branch and set

$$N = \frac{Y/p}{W_0(e^d Y/p)}. \quad (10.3)$$

Then $pN \log N + bN = Y$: setting $z = e^d N$ reduces the equation to $z \log z = e^d Y/p$. Let $t = t(y)$ be the unique sufficiently large solution of

$$g(t) = pt \log t + bt + \beta \log t = Y. \quad (10.4)$$

Its derivative is $p(\log t + 1 + d) + \beta/t > 0$ for large t , and

$$t = N - \frac{\beta \log N}{p(\log N + 1 + d)} + \mathcal{O}\left(\frac{1}{N \log N}\right). \quad (10.5)$$

Indeed the displayed displacement is $\mathcal{O}(1)$ and cancels $\beta \log N$ against the linear term of $pt \log t + bt$. The remaining residual is $\mathcal{O}(1/N)$, and division by a derivative of order $\log N$ proves the error. When $\beta = 0$, $t = N$ exactly.

Theorem 10.1 (Integer first-crossing bracket). *For the fixed ray define $\nu(y) = \min\{n \geq 1 : a_n \geq y\}$. There are constants $K, y_0 > 0$, depending on p, q , such that for $y \geq y_0$, with*

$$\eta(y) = \frac{K}{t(y)(1 + \log t(y))},$$

we have

$$\lceil t(y) - \eta(y) \rceil \leq \nu(y) \leq \lceil t(y) + \eta(y) \rceil. \quad (10.6)$$

The bracket eventually contains at most two adjacent integer candidates.

Proof. By (10.2), $|\log(a_n/D) - g(n)| \leq K_0/n$ beyond a fixed index. On a unit neighborhood of t , g' is bounded below by a positive multiple of $1 + \log t$. For sufficiently large K , the changes of g from t to $t \pm \eta$ dominate the possible K_0/n errors. More explicitly, the integer $\lceil t - \eta \rceil - 1$ is strictly below the target and $\lceil t + \eta \rceil$ is at or above it, by this derivative bound; both are within a fixed neighborhood of t . Eventual monotonicity then bounds every other large integer. Finally y eventually exceeds every value in the finite initial segment. This proves the first-crossing assertion, including equality targets. Since $2\eta \rightarrow 0$, the two ceiling endpoints differ by at most one eventually. $\square$

The constants and onset in this theorem are asymptotic existence constants. The statement alone is not a certified finite-input algorithm. Near an exact threshold, exact comparison of the remaining integer candidates is necessary; there is no justified unconditional single-ceiling rule. The uncertainty in (10.5) is a real interval before rounding, not a vanishing real error for the integer $\nu(y)$.

10.1 Higher-order inverse above the transition

For $\lambda > L$, let

$$\ell_j(\lambda) = \lceil u^j \rceil \log \left(\sum_{r \geq 0} c_r(\lambda) u^r \right).$$

For each fixed M ,

$$\log(a_n/D) = pn \log n + bn + \sum_{j=1}^M \frac{\ell_j(\lambda)}{q^j n^j} + \mathcal{O}_M(n^{-M-1}). \quad (10.7)$$

Let t_M solve the truncated smooth equation at target Y . The same proof gives a ceiling bracket around t_M of half-width

$$\mathcal{O}_M \left(\frac{t_M^{-M-1}}{1 + \log t_M} \right),$$

with fixed sufficiently large implicit constants. For $M = 1$, put $\alpha_1 = c_1(\lambda)/q$. The exact root of the first-order truncated equation satisfies

$$t_1 = N - \frac{\alpha_1}{pN(\log N + 1 + d)} + \mathcal{O} \left(\frac{N^{-3}}{(\log N)^2} \right). \quad (10.8)$$

Substitution leaves a residual $\mathcal{O}(N^{-3}/\log N)$ in that truncated equation. Its derivative is of order $\log N$, proving the displayed error. The discrete bracket has the larger half-width $\mathcal{O}(N^{-2}/\log N)$ because the forward logarithmic remainder is $\mathcal{O}(n^{-2})$.

Rational rays keep the dimensions integral without a changing fractional-part correction. These inverse assertions do not silently cover arbitrary sequences $w = \lfloor \lambda h \rfloor$, nor a nominal critical ray. Those require separate treatment of their lattice oscillations.

A Exact fixed height spectral structure

This appendix concerns an exact family already studied by Kotěšovec and Alekseyev. In particular, Kotěšovec's Conjecture 2 on p. 83, dated 15 September 2011, explicitly proposes maximal irreducible degree $\lfloor h/2 \rfloor + 1$ for the planar and horizontally cylindrical families; the accompanying discussion records cylindrical checks through $h = 20$ by Alekseyev [1, pp. 82–83]. The following argument proves the cylindrical upper bound and equality at infinitely many heights. It does not prove equality for every height or the planar conjecture, and does not establish whether these consequences were already proved elsewhere.

[write] 6 October 2026. The planar side is Part III (Report 175, written later the same day with this Part's Gram machinery). It proves the upper bound $\lfloor h/2 \rfloor + 1$ of Conjecture 2 for the planar family (Theorem 22.1) and Conjecture 4 (Corollary 22.2), and it records that Conjecture 3 as printed contradicts the exponents on the same page (Remark 25.1). Equality at every height remains open on both boards (Section 31, item 5). The write did not read the book; the dates and page numbers here are this Part's.

For fixed $h \geq 1$ write

$$\mathcal{F}_h(z) = \sum_{w \geq 1} A(h, w) z^w.$$

The Gram representation (3.2) gives a finite set $\mathcal{R}_h$ of distinct positive real algebraic integers and positive integer weights $a_h(\rho)$ with

$$A(h, w) = \sum_{\rho \in \mathcal{R}_h} a_h(\rho) \rho^w, \quad \mathcal{F}_h(z) = \sum_{\rho \in \mathcal{R}_h} a_h(\rho) \frac{\rho z}{1 - \rho z}. \quad (\text{A.1})$$

Complementing the word swaps the two colors and transposes C_b up to reorderings. Its nonzero singular-value squares and multiplicities are unchanged. No word equals its bitwise complement when $h \geq 1$, so every weight $a_h(\rho)$ is a positive even integer.

Each ρ produces a nonzero simple pole at $z = 1/\rho$, and all other terms are analytic there. Thus no spectral parameter cancels, and the minimal denominator is squarefree over characteristic zero. This proves the assertion recorded explicitly as Kotěšovec's Conjecture 1, dated 29 August 2011, on p. 83 [1]; the source records Alekseyev's checks through $h = 20$. We do not claim this is its first proof. The characteristic polynomial of each integer Gram matrix is integral, so algebraic conjugates occur with equal multiplicity. Summing over words shows that $a_h(\rho)$ is constant on each conjugacy class. Every conjugate is a nonnegative real Gram eigenvalue; nonzero conjugates are strictly positive. These observations explain the simple-factor and equal-coefficient patterns discussed on pp. 178–180 of the book; no new-priority claim is attached to that explanation.

If b has n_0 zero edges and n_1 one edges, then C_b has size $(n_1 + 1) \times (n_0 + 1)$. Either Gram matrix has the same nonzero spectrum. Choosing the smaller one proves

$$\deg_{\mathbb{Q}}(\rho) \leq \min(n_0 + 1, n_1 + 1) \leq \left\lfloor \frac{h+2}{2} \right\rfloor = \left\lfloor \frac{h}{2} \right\rfloor + 1. \quad (\text{A.2})$$

Hence every irreducible factor of the minimal fixed-height recurrence polynomial, equivalently the reversed minimal generating-function denominator, has degree at most this bound. It is stronger than the $h + 1$ bound credited to Alekseyev on p. 178, and matches the upper-bound portion of the earlier explicit conjecture on p. 83. The comparison to both passages is essential.

The bound is attained for infinitely many heights. Take $h = p - 3$, where $p \geq 5$ is an odd prime, and choose an alternating word. Its tree is the path on $p - 1$ vertices, whose largest squared adjacency eigenvalue is

$$\rho = 4 \cos^2 \left(\frac{\pi}{p} \right) = 2 + 2 \cos \left(\frac{2\pi}{p} \right).$$

For a primitive p th root of unity ζ , $[\mathbb{Q}(\zeta) : \mathbb{Q}] = p - 1$ and ζ satisfies $X^2 - (\zeta + \zeta^{-1})X + 1$ over the real subfield. Since ζ is nonreal, that quadratic extension has degree exactly two. Thus

$$\deg_{\mathbb{Q}}(\rho) = \frac{p-1}{2} = \left\lfloor \frac{h+2}{2} \right\rfloor.$$

The positive spectral weight prevents cancellation of this irreducible factor. Infinitely many primes give infinitely many equality heights.

For context, the fixed-height leading law is also transparent. Every C_b has $h + 1$ entries equal to one, so its squared norm is at most $h + 1$. Equality forces rank one, since its squared singular values sum to $h + 1$. A connected zero-one rank-one incidence graph is complete bipartite, and a complete bipartite tree is a star. Here that means b is constant. The two constant words each give the eigenvalue $h + 1$ once; every nonconstant word has smaller top eigenvalue. Therefore, for fixed h ,

$$A(h, w) \sim 2(h + 1)^w \quad (w \rightarrow \infty),$$

recovering the prior law on p. 192. This limit holds in a different parameter regime from Theorems 1.2 and 1.3.

[write] 6 October 2026. The same two facts (the Gram trace and the $h + 1$ unit entries of C_b) bound every word's contribution by $(h + 1)^w$, which gives $2(h + 1)^w \leq A(h, w) \leq 2^h(h + 1)^w$ for all $h, w \geq 1$, with the cases of equality: Remark 20.1 in Part II.

B Small cases and independent verification

The first exact rectangle is

$h \backslash w$	1	2	3	4	5	6
1	4	8	16	32	64	128
2	12	32	90	256	732	2102
3	32	100	344	1220	4392	15988
4	80	276	1082	4460	18890	81606
5	192	708	3036	13932	66532	327192
6	448	1732	7918	39316	205628	1118398

For example $A(1, 2) = 8$ but $A(2, 1) = 12$. The diagonal is 4, 32, 344, 4460, 66532, 1118398, in agreement with A137432.

[write] **6 October 2026.** Rows of this table are OEIS entries [3]: the row $h = 2$ is A194644 and the row $h = 5$ is A194647 (the first six terms of each agree with the entries); A195656 is the row $h = 12$, whose first term $53248 = 13 \cdot 2^{12}$ is $A(12, 1)$ by (B.1). The column $w = 1$ is also Part III's $P_h(1) = (h + 1)2^h$ (Lemma 23.1): with two block columns the cyclic and the planar boards have the same attacks.

Several formulas can be derived without asymptotic arguments:

$$A(h, 1) = (h + 1)2^h, \quad (\text{B.1})$$

$$A(1, w) = 2^{w+1}, \quad (\text{B.2})$$

$$A(h, 2) = (5h - 3)2^h + 4, \quad (\text{B.3})$$

$$A(2, w) = 2 \left[3^w + \left(\frac{3 + \sqrt{5}}{2} \right)^w + \left(\frac{3 - \sqrt{5}}{2} \right)^w \right]. \quad (\text{B.4})$$

For (B.1), choose h nonadjacent occupied rows from a path of $2h$ rows, in $h + 1$ ways, and choose either column for each king. For (B.2), the occupied columns alternate around the even cycle in two ways, and each of the w kings can use either row. For (B.3), the trace $\text{tr}(M_b^2)$ counts ordered threshold pairs. An unequal pair $u < v$ is allowed in both directions precisely when its interval of $v - u$ letters is constant. Thus

$$A(h, 2) = 2^h(h + 1) + 4 \sum_{j=1}^h (h + 1 - j)2^{h-j} = (5h - 3)2^h + 4.$$

For (B.4), two constant words contribute 3^w each; the other two have spectrum $0, (3 + \sqrt{5})/2, (3 - \sqrt{5})/2$. The conjugate power sum is also computable entirely in integers by $s_0 = 2, s_1 = 3, s_w = 3s_{w-1} - s_{w-2}$.

B.1 What the companion checks establish

The package contains independently written standard-library verification code. The direct geometric method enumerates legal masks in a physical row of cyclic width $2w$, requiring no adjacent cyclic bits. A mask s is compatible with a previous mask p when p avoids s and both cyclic one-step shifts of s . Dynamic programming through $2h$ rows tracks the largest king number and the number of ways attaining it. This method does not use the binary-run or tree encoding.

A separate implementation constructs every M_b directly from monotonicity of threshold intervals and sums exact integer traces. The companion compares these methods on a bounded rectangle, adds direct subset enumeration on tiny boards, checks $M_b = E_0 E_1$, and verifies the Gram/tree trace identity on bounded instances. Marked-run multiplicities and the summed first branch moment

are checked by exhaustive finite-word enumeration. Exact symbolic arithmetic checks the first two supercritical corrections and the critical Taylor and endpoint identities. These are finite checks supporting the derivation, not a proof of its asymptotic estimates.

Star-sum residuals are explicitly labeled as diagnostics for $S(h, w)$ or its base critical sum. A real formal exponent $w = Lh$ may be used there to isolate coefficient arithmetic; it is never presented as an exact king count. In particular a diagnostic convergence to $J(0) - 4$ tests the star constant only. The true count requires the separately proved spectral correction $4LG(0) = 4$.

The README states bounded default and extended modes, prerequisites, output paths, and limitations. The build and verification commands run in a fresh output directory, test the code with normal Python and with optimization enabled, compile the report, and record and verify file hashes. The source package uses an explicit allowlist; downloaded books, papers, and research working files are not redistributed. Numerical residual checks do not constitute interval-certified error constants.

C Sources scope and further questions

The principal exact-family source is Kotěšovec’s sixth edition [1], particularly pp. 82–83 and Section 2.6, pp. 169–196. Its p. 169 explicitly fixes the open dimension and lets the wrapped dimension grow; pp. 169–192 display exact formulas, recurrence and generating-function factorizations, and spectral power sums. Page 192 gives the fixed-height leading equivalent, and p. 193 tabulates leading and subleading roots. Page 178 credits Alekseyev’s September 2011 factor-generation method and a degree bound, while the earlier p. 83 gives the sharper explicit conjecture discussed in Appendix A. The linked Zealint discussion was not recovered in the bounded review [6]; its unavailability is a gap in historical coverage, not evidence about its contents or about priority.

Matsuo supplies the binary-word and cyclic-height encoding and an $\mathcal{O}(n^4)$ multiprecision-operation algorithm in the square setting [4]. Wilf’s earlier planar transfer-matrix analysis provides a related block-decomposition antecedent [8]. Planar maximum-density asymptotics in Wilf and Larsen [8, 9], and the later planar enumeration of Brown [10], have different boundary conditions. They should not be identified with the cylindrical joint-limit theorem. Likewise a cylinder problem with exactly one king in each row and column is a different density constraint.

The aspect-ratio transition and joint-limit laws in this report were not located in the primary materials inspected. This is a bounded comparison, not an exhaustive priority certification. Exact fixed-height enumeration, the square leading constant, and broad transfer-matrix ideas are established antecedents. The structural appendix is framed as a consequence of the exact representation with explicit credit to the prior conjecture, not as a claim of a newly proposed bound. No external publication, OEIS edit, author contact, or public repository modification is part of this report.

Several questions remain beyond the theorems proved here:

1. Determine the intermediate regimes as $\lambda_h \rightarrow 0$. The present uniform theorem requires a positive lower bound on w/h , and its one-macroscopic-run argument does not by itself cover the transition to fixed or slowly growing width.
2. Extend the critical expansion with effective uniform remainders to growing $|t_h|$, and determine maximal matching ranges with the two fixed-regime expansions. The large- $|t|$ formulas for G alone do not answer this.
3. Make the error constants and onsets effective, especially for the integer inverse brackets. This would turn the asymptotic brackets into certified finite-input threshold tools.

4. Derive subcritical corrections beyond the relative $\mathcal{O}(h^{-1})$ equivalent. The marked-star sum is explicit, but averaging the spectral perturbation with a long, extensive boundary needs additional coefficient analysis.
5. Resolve the exact maximal degree at every cylindrical height and locate any earlier complete proof of the 2011 conjecture. The infinitely many equality heights proved here do not settle the every-height assertion or its planar counterpart.

[write] **6 October 2026.** In the merged report: item 5's "planar counterpart" is advanced by Part III, which proves the planar upper bound but not attainment; the question about rectangular cylinders asked at the end of Part II (Section 21) is the subject of this Part. All five items are restated, with their status, in Section 31 (items 1, 2, 3, 4, 5 and 10).

Part II

Maximum density kings on a cylinder: all fixed order asymptotics and inverse thresholds for A137432

Bundle Report 173, archive `Cylindrical_Kings_All_Order_Asymptotics_and_Inverses_Source.zip`, delivered as `Report173.tex` (17 pages). Delivered title block “Maximum density kings on a cylinder / All fixed order asymptotics and inverse thresholds for A137432 / Report 173 / 3 October 2026” (empty author line). Printed as delivered, with `[write]` notes; Report 173’s Section k is Section $k + 10$ here, and its statements and equations move with their sections. Remark 20.1 is added in the write.

Abstract of this Part (as delivered). Let a_n count placements of n^2 mutually nonattacking kings on a labeled $2n \times 2n$ chessboard with horizontal wraparound and no vertical wraparound. An exact transfer matrix factors through a bipartite caterpillar tree. Its dominant contribution comes from a single run of length $n - \mathcal{O}(1)$ in a binary encoding of the placement. We prove, for every fixed M , an expansion $a_n = Cn^n(\sum_{j=0}^M c_j n^{-j} + \mathcal{O}_M(n^{-M-1}))$, with all $c_j \in \mathbb{Q}(e)$, and give a finite coefficient algorithm and exact c_1, c_2, c_3 . The leading equivalent and constant $C = 2e(e-1)^2/(e-2)^2$ were already recorded by Kotěšovec; the binary and height encoding and a polynomial-time exact algorithm are due to Matsuo. Further results include a total variation boundary law with rate $\mathcal{O}(n^{-1})$, the same rate for every fixed defect moment, a refined growth ratio, and a rounded inverse bracket. The package supplies independently written exact checks and diagnostic residuals. It is an unrefereed mathematical report, with bounded source comparisons and no exhaustive priority claim.

Reading conventions of Part II ([write])

n, a_n	The board is $2n \times 2n$, columns cyclic; $a_n = A(n, n)$ of Part I, OEIS A137432, $a_0 = 1$.
C, c_j, q	$C = C(1)$ and $c_j = c_j(1)$ of Part I; $q = e^{-1}$. The heights c_j of (12.2) are thresholds, not coefficients.
h, d	$h = k + 1$ in Section 15 (not a height); $d = 1 - k$, and in Section 16 also the number of outward runs.
L, K	$L = \log(x/C)$ in Section 18 (not $\log 2$); K the random boundary total, and $K = K_M(n)$ a cutoff.
G_i, H, R, S	$G_i(z)$ are Part I’s resolvents $R_i(z)$; H is Part I’s $\mathcal{H}_1$; R, S are incidence matrices (Part I’s R_0, R_1), and $S = s_L + s_R$ is also a branch moment.

11 Main results and conventions

The board consists of squares (r, s) with $0 \leq r, s < 2n$. Columns are read modulo $2n$; rows are not identified. Two distinct kings attack when their row difference is at most one and their circular column distance is at most one. We count placements on this labeled board, without quotienting by rotations, reflections, or translations. Set $a_0 = 1$. For $n \geq 1$ the sequence is [OEIS A137432](#); it begins

4, 32, 344, 4460, 66532, 1118398, 20984924, . . .

Partitioning the board into disjoint 2×2 blocks gives an upper bound of n^2 kings. Occupying every other row and every other column attains it. Thus the count in this report is exactly a maximum-density count.

Theorem 11.1 (All fixed orders). *Put*

$$C = \frac{2e(e-1)^2}{(e-2)^2} = 31.1116835720490503682643922791 \dots \quad (11.1)$$

There is a sequence $c_0, c_1, \dots \in \mathbb{Q}(e)$, with $c_0 = 1$, such that for every fixed integer $M \geq 0$,

$$a_n = Cn^n \left(\sum_{j=0}^M \frac{c_j}{n^j} + \mathcal{O}_M(n^{-M-1}) \right) \quad (n \rightarrow \infty). \quad (11.2)$$

Every coefficient is computable by a finite rational-function calculation. The first three nonconstant coefficients are given in Section 16.

The quantifier “fixed M ” is essential. We assert neither convergence of the infinite asymptotic series nor a remainder bound uniform in growing M . No fitted coefficients enter the proof.

[write] **6 October 2026.** Second route. This theorem is Part I’s Theorem 1.2 at $\lambda_h = 1$, proved there for every compact supercritical range of w/h (Part I was written after this Part, the same day). The two proofs are independent paraphrases of one argument (localization at one long run, the Schur equation, a logarithmic cutoff, an analytic disk and Cauchy estimates; Sections 14–16 here, Section 6 there); this Part’s disk has explicit radii. The coefficients agree: Part I’s $c_1(1)$, $c_2(1)$ are (16.5), (16.6) (checked exactly by the write; note at the end of Section 1), and by uniqueness of the expansion $c_j = c_j(1)$ for every j . The exact c_3 (16.7) is in this Part only. The leading constant is Kotěšovec’s (front matter).

Theorem 11.2 (Boundary law). *Choose a placement uniformly from the a_n possibilities and form the binary word described in Section 12. Except on an event of exponentially small probability, this word has a unique run longer than $n/2$. Let its central bit and its left and right outward run compositions be the associated state. Write K for the total length of the two boundaries; on the exceptional event define $K = n -$ the maximum run length.*

Let $q = e^{-1}$ and $F(q) = (1-q)/(1-2q)$. The state converges in total variation, at rate $\mathcal{O}(n^{-1})$, to a uniform bit independent of two independent finite compositions. A specified composition of total length l has probability $q^l/F(q)$. The exceptional state has limiting probability zero. Moreover, for every fixed integer $p \geq 0$,

$$\mathbb{E}[K^p] = \mathbb{E}[K_\infty^p] + \mathcal{O}_p(n^{-1}), \quad \mathbb{E}[z^{K_\infty}] = \frac{F(qz)^2}{F(q)^2} \quad (|z| \leq 1). \quad (11.3)$$

In particular $\mathbb{E}[K] = 2e/((e-1)(e-2)) + \mathcal{O}(n^{-1})$.

Independence refers to run compositions, or to binary boundaries encoded relative to the central bit. The literal inner boundary bit must be opposite to the central bit and is therefore not independent of it.

Corollary 11.3 (Ratio and inverse). *The sequence is eventually strictly increasing and*

$$\frac{a_{n+1}}{na_n} = e \left(1 + \frac{1}{2n} + \frac{-c_1 - 1/24}{n^2} + \mathcal{O}(n^{-3}) \right). \quad (11.4)$$

Write $\ell_j = [u^j] \log(\sum_{r \geq 0} c_r u^r)$, and

$$g_M(t) = t \log t + \sum_{j=1}^M \ell_j t^{-j}.$$

For sufficiently large x , let $t_M(x)$ be the unique sufficiently large root of $g_M(t) = \log(x/C)$. There is a constant $A_M > 0$ such that, with

$$\eta_M(x) = \frac{A_M t_M(x)^{-M-1}}{1 + \log t_M(x)}, \quad \nu(x) = \min\{n \geq 0 : a_n \geq x\},$$

we have the precise integer bracket

$$\lceil t_M(x) - \eta_M(x) \rceil \leq \nu(x) \leq \lceil t_M(x) + \eta_M(x) \rceil. \quad (11.5)$$

The bracket eventually contains at most two adjacent integers. Its constants and onset are asymptotic existence constants, not certified finite-input numerical bounds.

[write] 6 October 2026. *The A137432 conjecture.* The entry A137432 [2] records “Conjecture: limit of $a(n+1)/(n^*a(n))$ as $n \rightarrow \infty$ is e ,” added by Vaclav Kotesovec on 31 August 2011 (revision #19, the entry’s first version under its present name). Its object is literally the a_n of this Part, and (11.4) proves it; the limit alone needs only $M = 0$ in Theorem 11.1, since $a_{n+1}/(na_n) = (1 + 1/n)^{n+1}(1 + \mathcal{O}(n^{-1}))$. The expansion (11.4) itself is in this Part only. Nothing was submitted to the OEIS.

Prior results and scope. Kotěšovec’s leading equivalent and the value of C are explicitly recorded in the OEIS entry, dated 29 July 2023 and updated 18 March 2024 [2]. Matsuo’s binary-word and cyclic-height encoding, and his $\mathcal{O}(n^4)$ multiprecision-operation algorithm, are existing work [4]. The report derives the encoding afresh to fix conventions and supplies a caterpillar reduction, uniform expansion, exact corrections, and boundary law. These results go beyond those displayed in the sources inspected here; no first-in-history or exhaustive novelty claim is made. Section 20 explains the source and verification boundaries.

12 From placements to a transfer matrix

Index the n^2 blocks by (i, j) , $0 \leq i, j < n$. The king in block (i, j) has coordinates

$$(2i + x_{ij}, 2j + y_{ij}), \quad x_{ij}, y_{ij} \in \{0, 1\}.$$

Every block contains exactly one king. Consider neighboring blocks in the same two-row band. If $y_{ij} = 1$ and $y_{i,j+1} = 0$, their column distance is one and their row distance is at most one, so they attack. Consequently the cyclic sequence of column offsets cannot descend. It is constant:

$$y_{ij} = b_i \quad (0 \leq j < n), \quad b_i \in \{0, 1\}. \quad (12.1)$$

This argument also covers the boundary between the last and first block columns. For $n = 1$ constancy is automatic.

In a fixed block column, an adjacent pair with $x_{ij} = 1$ and $x_{i+1,j} = 0$ would similarly attack. Hence x_{ij} is nondecreasing in i , and there is a unique height $c_j \in \{0, \dots, n\}$ with

$$x_{ij} = \begin{cases} 0, & i < c_j, \\ 1, & i \geq c_j. \end{cases} \quad (12.2)$$

Only conflicts between diagonally neighboring blocks remain. Put $u = c_j$ and $v = c_{j+1}$, with block-column indices cyclic. A down-right attack from (i, j) to $(i+1, j+1)$ occurs exactly when

$$u \leq i < v - 1, \quad (b_i, b_{i+1}) = (1, 0).$$

Thus, when $u < v$, the substring $b_u, \dots, b_{v-1}$ must be nondecreasing. An attack in the opposite diagonal direction occurs exactly when

$$v \leq i < u - 1, \quad (b_i, b_{i+1}) = (0, 1),$$

so when $u > v$, the substring $b_v, \dots, b_{u-1}$ must be nonincreasing. Equal heights always work. These conditions are sufficient as well: every possibly attacking pair lies in the same block or in horizontally, vertically, or diagonally adjacent blocks, all of which have been checked. For small widths, two directions can meet the same neighboring block; the cyclic conditions check both directions.

For a binary word $b = (b_0, \dots, b_{n-1})$ let M_b be the $(n+1) \times (n+1)$ matrix indexed by $0, \dots, n$, with

$$M_b[u, v] = \begin{cases} 1, & u = v, \\ 1, & u < v \text{ and } b_u \cdots b_{v-1} \text{ has the form } 0^*1^*, \\ 1, & u > v \text{ and } b_v \cdots b_{u-1} \text{ has the form } 1^*0^*, \\ 0, & \text{otherwise.} \end{cases} \quad (12.3)$$

Here a starred block may be empty. For fixed b , a permitted list $(c_0, \dots, c_{n-1})$ contributes one to the product $\prod_j M_b[c_j, c_{j+1}]$, where $c_n = c_0$. Therefore

$$a_n = \sum_{b \in \{0,1\}^n} \text{tr}(M_b^n). \quad (12.4)$$

The trace sums over labeled cyclic lists with a distinguished position 0. We do not divide by n or by a rotational stabilizer. This is precisely the labeled-board convention.

Remark 12.1. Matsuo's presentation contains the same binary and height constraints [4]. The explanatory row and column variable names appear transposed there. Equations (12.1)–(12.2) specify our coordinates directly and avoid depending on that naming convention.

[write] **6 October 2026.** A presentational remark about Matsuo's README; the write did not read that README, and no statement here depends on it.

13 Component factorization and the caterpillar

Color the edges of the path on vertices $0, \dots, n$ by $b_0, \dots, b_{n-1}$. Let $E_\epsilon[u, v] = 1$ if every edge between u and v has color ϵ , for $\epsilon = 0, 1$. Empty intervals qualify. Thus E_ϵ is the matrix of membership in a common monochromatic component, including isolated vertices.

Lemma 13.1. *The exact matrix identity $M_b = E_0 E_1$ holds.*

Proof. An intermediate vertex w counted by $(E_0 E_1)[u, v]$ must lie in the same 0-component as u and the same 1-component as v . It cannot lie outside the interval with endpoints u, v : a nonempty subinterval would then have to consist entirely of both colors. There cannot be two possible vertices w , for the interval between them would have the same contradiction. For $u < v$, existence of w says exactly that the interval has form 0^*1^* ; for $u > v$ its increasing-order form is 1^*0^* . For $u = v$ the unique choice is $w = u$. $\square$

Let R and S be the vertex-component incidence matrices for the 0-components and 1-components. Then

$$E_0 = R R^\top, \quad E_1 = S S^\top, \quad C_b = R^\top S.$$

Two opposite-color components intersect in at most one path vertex, so C_b has entries zero or one. Form a bipartite graph T_b on these components, giving one edge for each nonempty intersection. Its $n + 1$ edges correspond exactly to the original path vertices. It is connected, since successive path vertices give edges sharing the component of their intervening color. There are $n + 2$ components altogether: if there are n_0 zero edges and n_1 one edges, their component counts are $n + 1 - n_0$ and $n + 1 - n_1$. Thus T_b is a connected graph with one fewer edge than vertices, hence a tree.

Cyclicity of matrix trace gives, for $n \geq 1$,

$$\text{tr}(M_b^n) = \text{tr}((C_b C_b^\top)^n), \quad (13.1)$$

$$A(T_b) = \begin{pmatrix} 0 & C_b \\ C_b^\top & 0 \end{pmatrix}, \quad \text{tr}(M_b^n) = \frac{1}{2} \text{tr}(A(T_b)^{2n}). \quad (13.2)$$

This proof does not require diagonalizing the nonsymmetric matrix M_b . The nonzero adjacency eigenvalues occur in pairs $\pm \sigma_j(C_b)$, and all eigenvalues contributing on the Gram side are nonnegative.

Lemma 13.2 (Run description). *If b has positive run lengths $(r_1, \dots, r_t)$, then T_b is a caterpillar. For $t \geq 2$ its spine has t vertices, with pendant leaf counts*

$$r_1, r_2 - 1, \dots, r_{t-1} - 1, r_t.$$

For $t = 1$ it is a star with $n + 1$ leaves. Every run of length r_i corresponds to a vertex of degree $r_i + 1$.

Proof. The nonsingleton monochromatic components are the runs. Consecutive runs intersect at their shared endpoint, producing the spine edges. At an internal run, its $r_i - 1$ interior path vertices give singleton components of the opposite color and hence leaves. An endpoint run has one additional free path endpoint, producing r_i leaves. If there is only one run, every path vertex meets an opposite-color singleton, giving $n + 1$ leaves. The degree statements follow immediately. $\square$

A composition of n is an ordered list of positive integers summing to n . Each composition is the run list of exactly two binary words. If T_r denotes its tree, (12.4) and (13.2) give the useful exact identity

$$a_n = \sum_{r \text{ a composition of } n} \text{tr}(A(T_r)^{2n}). \quad (13.3)$$

The factor two from the initial bit has canceled the factor one-half in the tree trace. This is a sum over ordered compositions, not over unlabeled tree isomorphism classes.

14 A dominant run and its spectral equation

Suppose the word has a run of length $m > n/2$. It is unique. Delete the corresponding hub from T_b and write

$$k = n - m = l + r,$$

where l, r are the total boundary lengths on its two sides. Each nonempty side has its nearest bit opposite to the central bit. Its outward run list is an arbitrary composition. If α_l is the number of compositions of l , with the empty composition counted once, then

$$\alpha_0 = 1, \quad \alpha_l = 2^{l-1} \ (l \geq 1), \quad F(q) = \sum_{l \geq 0} \alpha_l q^l = \frac{1 - q}{1 - 2q}. \quad (14.1)$$

There are $2\alpha_l\alpha_r$ such binary words for fixed l, r . The order on the left boundary is outward from the hub, not left to right on the original path.

Let A_L, A_R be the adjacency matrices of the nonempty rooted branches after deleting the hub, with roots v_L, v_R . Besides them, the deletion leaves isolated leaves. The total number of branch edges is k , since the original tree has $n + 1$ edges and the removed hub has degree $m + 1$. Empty branches contribute zero throughout.

Lemma 14.1 (Schur complement). *Let Λ be the square of the Perron adjacency eigenvalue of T_b . Then $\Lambda \geq m + 1 > k$, and*

$$\Lambda = m + 1 + \sum_{j \geq 1} B_j \Lambda^{-j}, \quad B_j = (A_L^{2j})[v_L, v_L] + (A_R^{2j})[v_R, v_R], \quad (14.2)$$

with $0 \leq B_j \leq k^j$ for $j \geq 1$.

Proof. The hub and its neighbors induce a star of degree $m + 1$, so the variational principle gives $\Lambda \geq m + 1$. A bipartite graph with K edges has squared adjacency norm at most K : the squared norm of its bipartite incidence matrix is at most its squared Frobenius norm, which is K . If the two branches have k_L, k_R edges, then their norms are at most $\sqrt{k_L}, \sqrt{k_R}$, strictly below $\sqrt{\Lambda}$.

Apply the Schur complement at $\lambda = \sqrt{\Lambda}$. Its diagonal resolvents have convergent Neumann series. A tree is bipartite, so diagonal odd powers vanish. Multiplication by λ gives the zeroth contribution equal to the number of hub neighbors, namely $m + 1$, followed by (14.2). Finally, $B_j \leq k_L^j + k_R^j \leq (k_L + k_R)^j = k^j$. $\square$

Write s_L, s_R for the first outward run lengths, t_L, t_R for the second, and u_L, u_R for the third, using zero for absent runs. Direct rooted closed-walk counting, or the resolvent recurrence in Section 16, yields

$$B_1 = s_L + s_R =: S, \quad (14.3)$$

$$B_2 = s_L^2 + t_L + s_R^2 + t_R =: T, \quad (14.4)$$

$$B_3 = \sum_{a \in \{L, R\}} (s_a^3 + 2s_a t_a + t_a^2 + u_a). \quad (14.5)$$

For example, the root degree is s_a . Four-step returns either choose two successive root edges, giving s_a^2 , or visit the next spine vertex and use one of its outward edges, giving t_a .

Lemma 14.2 (Spectral remainder). *For every such word,*

$$\text{tr}(M_b^n) = \Lambda^n + R_b, \quad 0 \leq R_b \leq nk^n. \quad (14.6)$$

Proof. The adjacency matrix after hub deletion has norm at most $\sqrt{k}$. By symmetric eigenvalue interlacing, at most one eigenvalue of $A(T_b)$ exceeds $\sqrt{k}$. Bipartite symmetry gives the corresponding statement below $-\sqrt{k}$. The tree is connected, so its Perron pair is simple. All remaining eigenvalues have absolute value at most $\sqrt{k}$. Their even powers contribute nonnegative terms to one-half of the trace; there are at most n remaining terms after division by two and the harmless coarse bound in (14.6) follows. $\square$

With $d = 1 - k$, expansion of (14.2) at fixed boundary data starts as

$$\Lambda = n + d + \frac{S}{n} + \frac{T - dS}{n^2} + \mathcal{O}_k(n^{-3}), \quad (14.7)$$

$$\left(\frac{\Lambda}{n}\right)^n = e^d \left(1 + \frac{P_1}{n} + \frac{P_2}{n^2} + \cdots\right), \quad (14.8)$$

$$P_1 = S - \frac{d^2}{2},$$

$$P_2 = T - 2dS + \frac{d^3}{3} + \frac{1}{2} \left(S - \frac{d^2}{2}\right)^2.$$

The pointwise expansion is only a starting point. The next section supplies the uniform bounds needed before summing over all boundaries.

15 Uniform tails and all fixed order remainders

15.1 Words with no almost full run

Let m be the maximum run length, without assuming $m > n/2$. Split the caterpillar adjacency matrix into pendant-star edges and spine-path edges. The first matrix is a disjoint union of stars with maximum degree at most $m + 1$; the second has norm at most two. The triangle inequality gives

$$\Lambda \leq (\sqrt{m+1} + 2)^2 = m + \mathcal{O}(\sqrt{n}). \quad (15.1)$$

The one-run star also satisfies this bound. There are at most $2n2^{n-m}$ words with maximum run length m : choose a marked start of one such run, its bit, and arbitrary remaining bits. The overcount is harmless. Also (13.1) implies

$$\text{tr}(M_b^n) \leq (n+1)\Lambda^n.$$

For $n/4 \leq m \leq (1-\epsilon)n$, set $t = m/n$. Apart from polynomial factors and an $\exp(\mathcal{O}(\sqrt{n}))$ factor from (15.1), the normalized exponential rate is

$$f(t) = (1-t) \log 2 + \log t.$$

On $(0, 1]$, $f'(t) = 1/t - \log 2 > 0$ and $f(1) = 0$. Thus f is uniformly negative on $[1/4, 1-\epsilon]$. For $m \leq n/4$, the crude count 2^n and (15.1) instead give exponential base $2(1/4 + o(1)) < 1$. Summing over m proves that, for every fixed $\epsilon \in (0, 1)$, some $c_\epsilon > 0$ satisfies

$$\sum_{b: \max \text{run}(b) \leq (1-\epsilon)n} \text{tr}(M_b^n) = \mathcal{O}(n^n e^{-c_\epsilon n}). \quad (15.2)$$

If the first interval is empty, only the second estimate is needed, or one can use any smaller positive ϵ .

15.2 Almost star bounds and a logarithmic cutoff

Fix $0 < \epsilon < 1/2$. For $k \leq \epsilon n$, Lemma 14.1 gives

$$0 \leq \Lambda - (n - k + 1) \leq \sum_{j \geq 1} (k/\Lambda)^j = \frac{k}{\Lambda - k} \leq \frac{k}{n - 2k + 1}. \quad (15.3)$$

Using $\log v \leq v - 1$ yields

$$\left(\frac{\Lambda}{n}\right)^n \leq \exp\left(1 - k + \frac{k}{n - 2k + 1}\right) \leq C_\epsilon e^{-k}. \quad (15.4)$$

There are at most $2(k+1)2^k$ central-bit and boundary-composition states of total length k . Put $\rho = 2/e < 1$. Their Perron contribution after normalization by n^n has tail bounded by

$$C_\epsilon \sum_{k>K} (k+1)\rho^k. \quad (15.5)$$

The trace remainder can be treated separately. By (14.6), its aggregate over $k \leq \epsilon n$ is bounded by a constant times

$$n \sum_{1 \leq k \leq \epsilon n} (k+1)2^k (k/n)^n = \mathcal{O}(n^3(\epsilon 2^\epsilon)^n).$$

Indeed $t \mapsto \log t + t \log 2$ is increasing for $t > 0$. Since $\epsilon < 1/2$, the base $\epsilon 2^\epsilon$ is less than one. Any fixed power of k can be inserted with only a polynomial change in the prefactor. This observation is also needed for the probability law.

The remaining words are exponentially small by (15.2). A cutoff

$$K = K_M(n) = A_M \log n$$

with sufficiently large fixed A_M consequently removes all omitted words at any specified algebraic order. We next show that the local Taylor remainder can be summed without a logarithmic loss.

15.3 A uniform analytic disk

Put $h = k + 1$, $d = 1 - k$, $u = 1/n$, and $y = \Lambda/n$. The exact implicit equation is

$$y = 1 + du + \sum_{j \geq 1} B_j u^{j+1} y^{-j}. \quad (15.6)$$

Regard the boundary data as fixed and u as complex. On

$$|u| \leq \frac{1}{16h}, \quad |y - 1| \leq \frac{1}{4},$$

set $r = k|u|/|y| \leq 1/12$. The correction sum on the right of (15.6) has absolute value at most

$$|u| \frac{r}{1-r} \leq \frac{|u|}{11},$$

and its derivative with respect to y has absolute value at most

$$\frac{|u|}{|y|} \frac{r}{(1-r)^2} \leq \frac{1}{121h}.$$

Together with $|du| \leq 1/16$, these inequalities show that the right side maps the closed y -disk strictly into itself and is a uniform contraction. Iterating from $y = 1$ produces its unique analytic fixed point. On a possibly smaller absolute disk of the same scale,

$$y = 1 + du + \mathcal{O}(ku^2), \quad \frac{\log y}{u} - d = \mathcal{O}(h^2|u|). \quad (15.7)$$

For $k = 0$ the first correction is exactly zero. The logarithm is its analytic branch near one, and its apparent pole at zero is removable.

For real $u = 1/n$ in the logarithmic window, (15.3) places the actual Perron solution inside this disk; uniqueness identifies it with the analytic solution. Now define

$$H(u) = \exp\left(\frac{\log y(u)}{u} - d\right) = \sum_{j \geq 0} P_j u^j. \quad (15.8)$$

The distinction between the two radii is important. The solution y is analytic on a disk of radius proportional to h^{-1} , but to bound H by an absolute constant we use the smaller disk

$$|u| \leq r_h := \frac{c}{h^2} \quad (15.9)$$

for a sufficiently small fixed $c > 0$. Equation (15.7) implies $|H(u)| \leq C$ there, with absolute C . Cauchy's coefficient estimates give

$$|P_j| \leq C_j h^{2j}, \quad \left| H(1/n) - \sum_{j=0}^M P_j n^{-j} \right| \leq C_M h^{2M+2} n^{-M-1}, \quad (15.10)$$

provided $1/n \leq r_h/2$. To see the remainder bound directly, bound the coefficient of order j by $C r_h^{-j}$ and sum the geometric tail from $M+1$ onward. This condition holds uniformly for $k \leq A_M \log n$ once n is large enough.

15.4 Summation and completion of the expansion proof

For a state with boundary total k , equations (15.8) and (15.10) say

$$\left(\frac{\Lambda}{n}\right)^n = e^{1-k} \left(\sum_{j=0}^M P_j n^{-j} + \mathcal{O}_M((k+1)^{2M+2} n^{-M-1}) \right).$$

Summing the error over the logarithmic window costs at most

$$C_M n^{-M-1} \sum_{k \geq 0} (k+1)^{2M+3} \rho^k = \mathcal{O}_M(n^{-M-1}).$$

Every polynomial moment of ρ^k is summable. The omitted Perron tail, bulk, and non-Perron part are smaller by the preceding bounds. The coefficient bound in (15.10) allows the logarithmic cutoff to be removed from each coefficient with the same precision, after increasing A_M if necessary.

Let $\mathcal{C}$ be the set of all finite compositions, including the empty one, and write $|\alpha|$ for their total length. Thus the coefficients are the absolutely convergent weighted sums

$$C c_j = 2e \sum_{\alpha, \beta \in \mathcal{C}} q^{|\alpha|+|\beta|} P_j(\alpha, \beta), \quad q = e^{-1}. \quad (15.11)$$

The factor two is the central bit. Since $P_0 = 1$, the leading sum is $2eF(q)^2$, exactly (11.1). Section 16 proves that each normalized coefficient lies in $\mathbb{Q}(e)$ and is finitely computable, completing Theorem 11.1.

16 Exact coefficients and a finite algorithm

16.1 Rooted resolvents

For a branch whose outward runs are $s_1, \dots, s_d$, define formal series

$$G_{d+1}(z) = 1, \quad G_i(z) = \frac{1}{1 - z(s_i - 1 + G_{i+1}(z))} \quad (i = d, \dots, 1). \quad (16.1)$$

The normalized diagonal resolvent at the root is $G_1(z)$. To verify this, remove the root. For an internal branch root there are $s_i - 1$ pendant leaves and one outward spine branch. Each leaf contributes one to the normalized self-energy, while the spine branch contributes G_{i+1} . At the terminal vertex there are s_d actual leaves, which agrees with the extra terminal value $G_{d+1} = 1$.

Thus

$$B_j = [z^j](G_{L,1}(z) + G_{R,1}(z) - 2), \quad j \geq 1.$$

An absent side has $G_1 = 1$. Padding missing terminal runs by zero preserves this value: $1/(1 - z(0 - 1 + 1)) = 1$. Repeated truncation of (16.1) shows that its coefficient of z^j is a polynomial in only the first j outward runs. Equations (14.3)–(14.5) follow by expansion.

16.2 Formal inversion and weighted summation

For a requested fixed order M , the following finite procedure computes $c_0, \dots, c_M$.

1. Truncate (16.1) at z^{M+1} on each side. Only the first M outward run variables are needed. Extract $B_1, \dots, B_M$.
2. Solve (15.6) recursively in the ring of formal power series modulo u^{M+2} , starting with $y = 1 + du$. The coefficient of u^r on the right depends only on previously determined coefficients.
3. Form $\log y/u - d$ modulo u^{M+1} and exponentiate. This produces the rational polynomials $P_0, \dots, P_M$ in k and the finitely many run variables.
4. Sum each monomial over the two finite compositions using the rules below, then substitute $q = e^{-1}$ and divide by $F(q)^2$.

There is no requirement to enumerate infinitely many compositions numerically.

Put

$$h_a(q) = \sum_{s \geq 1} s^a q^s, \quad D = q \frac{d}{dq}.$$

Here $h_0 = q/(1 - q)$ and $h_a = D^a h_0$ are rational functions. If a monomial in the initial outward runs is $s_1^{a_1} \dots s_d^{a_d}$ with $a_d > 0$, then its sum over all finite compositions is

$$F(q) \prod_{i=1}^d h_{a_i}(q). \quad (16.2)$$

Indeed the first d positive lengths are chosen independently, and the remaining composition is arbitrary. Required but unweighted earlier lengths contribute h_0 ; compositions shorter than d contribute zero. The constant monomial has sum $F(q)$. The two sides multiply, and multiplying a summand by k^p is implemented by D^p on the joint weight. This last statement refers to the total boundary length, not merely to the displayed runs. Every operation is rational over $\mathbb{Q}(q)$. Consequently (15.11) gives $c_j \in \mathbb{Q}(q) = \mathbb{Q}(e)$ for all j .

16.3 The first three corrections

For convenient verification define $d = 1 - k$ and

$$\begin{aligned} E_1 &= B_1 - \frac{1}{2}d^2, \\ E_2 &= B_2 - 2dB_1 + \frac{1}{3}d^3, \\ E_3 &= B_3 - 3dB_2 + 3d^2B_1 - \frac{3}{2}B_1^2 - \frac{1}{4}d^4. \end{aligned}$$

These are the first three coefficients of $\log y/u - d$. Hence

$$P_1 = E_1, \quad P_2 = E_2 + \frac{1}{2}E_1^2, \quad P_3 = E_3 + E_1E_2 + \frac{1}{6}E_1^3. \quad (16.3)$$

Applying (16.2) gives the following exact expressions. To keep their polynomial numerators legible, put $E = e$ and write

$$\begin{aligned} N_1(E) &= E^4 - 12E^3 + 45E^2 - 40E + 4, \\ N_2(E) &= 11E^8 - 82E^7 + 694E^6 - 4902E^5 + 19179E^4 \\ &\quad - 34836E^3 + 28984E^2 - 9200E + 176, \\ N_3(E) &= 21E^{12} - 484E^{11} - 17E^{10} + 37848E^9 - 268609E^8 \\ &\quad + 850396E^7 - 1209231E^6 + 86272E^5 + 2100764E^4 \\ &\quad - 2882112E^3 + 1642864E^2 - 358912E + 1344. \end{aligned} \quad (16.4)$$

Then

$$c_1 = -\frac{N_1(e)}{2(e-2)^2(e-1)^2} = -13.5720521590463861937039393882\dots, \quad (16.5)$$

$$c_2 = \frac{N_2(e)}{24(e-2)^4(e-1)^4} = 576.331755295050712336706653454\dots, \quad (16.6)$$

$$c_3 = -\frac{N_3(e)}{48(e-2)^6(e-1)^6} = -32709.3531657894628802177731924\dots \quad (16.7)$$

The growing early coefficients explain why moderate- n numerical residuals approach their limits slowly. They do not justify claims about the behavior of c_j as $j \rightarrow \infty$.

As a shorter algebraic cross-check for the first two orders, set

$$W = F^2, \quad W_S = 2F^2h_1, \quad W_{S^2} = 2F^2h_2 + 2F^2h_1^2, \quad W_T = 2F^2(h_2 + h_0h_1), \quad \delta = 1 - D.$$

Here δ is an operator and powers mean repeated application. Then

$$\begin{aligned} c_1 &= \frac{W_S - \delta^2 W/2}{W}, \\ c_2 &= \frac{W_T - 2\delta W_S - \delta^2 W_S/2 + W_{S^2}/2 + \delta^3 W/3 + \delta^4 W/8}{W}, \end{aligned} \quad (16.8)$$

evaluated at $q = e^{-1}$. This independently organizes the same rational-function moments.

17 Boundary distribution with rates

Let Ω be the countable set of a specified central bit and two finite compositions, together with an exceptional symbol $\star$. For ordinary ω let $k(\omega)$ be its total composition length. If $k < n/2$, the corresponding word of length n is uniquely determined by inserting a central run of length $n - k$. Define its unnormalized placement mass, divided by n^n , as

$$w_n(\omega) = \frac{\text{tr}(M_b^n)}{n^n}.$$

For ordinary states with $k \geq n/2$, set $w_n(\omega) = 0$. Let $w_n(\star)$ be the total normalized mass of words with no run longer than $n/2$. The limiting unnormalized masses are

$$w_\infty(\omega) = e^{1-k(\omega)} \quad (\omega \neq \star), \quad w_\infty(\star) = 0. \quad (17.1)$$

Their total is C .

Proposition 17.1 (Weighted mass convergence). *For every fixed integer $p \geq 0$,*

$$\sum_{\omega \neq \star} (1 + k(\omega))^p |w_n(\omega) - w_\infty(\omega)| + (n + 1)^p w_n(\star) = \mathcal{O}_p(n^{-1}). \quad (17.2)$$

Proof. First remove the non-Perron contribution over $k \leq \epsilon n$ by the exponentially small estimate following (15.5). Its weighted sum is still exponentially small. The bulk estimate (15.2) handles all other finite- n words outside this range, even after multiplying by $(n + 1)^p$. In particular it handles $\star$.

On $k \leq A \log n$, (15.10) with $M = 0$ gives

$$\left| \left(\frac{\Lambda}{n} \right)^n - e^{1-k} \right| \leq \frac{C e^{1-k} (k + 1)^2}{n}.$$

After inserting the number of states and the weight $(1 + k)^p$, its sum is $\mathcal{O}_p(n^{-1})$. In $A \log n < k \leq \epsilon n$, both Perron and limiting masses have weighted tails bounded by a constant times

$$\sum_{k > A \log n} (k + 1)^{p+1} \rho^k.$$

Choose A sufficiently large to make this $\mathcal{O}_p(n^{-2})$. Finally, limiting states with $k \geq \epsilon n$, including those absent at finite n , have exponentially small weighted tails. Combining these estimates proves (17.2). $\square$

The true total mass is $a_n/n^n = C + \mathcal{O}(n^{-1})$. Divide (17.2) by it and use the finite limiting weighted moments to account for normalization. For $p = 0$, one-half the resulting ℓ^1 bound proves the asserted total variation rate. For each fixed p , the ordinary-state statistic $K^p = k^p$ and the exceptional bound $K^p \leq n^p$ prove (11.3).

The normalized limiting mass at one specified bit and composition pair is

$$\frac{e^{1-k}}{C} = \frac{q^k}{2F(q)^2}. \quad (17.3)$$

This factorization proves the uniform-bit and independence assertions. For one boundary let R be its number of runs and put $h = q/(1 - q) < 1$. Since a positive run has total weight h , we have

$$\mathbb{P}(R = r) = (1 - h)h^r \quad (r \geq 0).$$

Conditional on R , the run lengths are independent and have distribution

$$\mathbb{P}(s = j) = (1 - q)q^{j-1} \quad (j \geq 1).$$

The empty boundary occurs when $R = 0$. The two-boundary probability generating function is $F(qz)^2/F(q)^2$, and differentiating at $z = 1$ gives

$$\mathbb{E}[K_\infty] = 2q \frac{F'(q)}{F(q)} = \frac{2q}{(1-q)(1-2q)} = \frac{2e}{(e-1)(e-2)}.$$

This completes Theorem 11.2 with its moment rates.

18 Growth ratio and inverse thresholds

18.1 Ratio expansion

Write $A(n) = a_n/(Cn^n)$. Theorem 11.1 implies

$$\frac{A(n+1)}{A(n)} = 1 - \frac{c_1}{n^2} + \mathcal{O}(n^{-3}).$$

Also

$$\left(1 + \frac{1}{n}\right)^{n+1} = e \left(1 + \frac{1}{2n} - \frac{1}{24n^2} + \mathcal{O}(n^{-3})\right).$$

Multiplying gives (11.4), including the previously recorded ratio-limit conjecture. Since $a_{n+1}/a_n \sim en$, eventual strict increase follows. Higher ratio corrections are obtained by ordinary formal multiplication of the fixed-order expansion; the uniform remainder follows by choosing one more fixed order than needed.

18.2 Continuous approximations to the threshold

Taking logarithms at fixed order gives

$$\log(a_n/C) = n \log n + \sum_{j=1}^M \ell_j n^{-j} + \mathcal{O}_M(n^{-M-1}), \quad (18.1)$$

where

$$\ell_1 = c_1, \quad \ell_2 = c_2 - \frac{1}{2}c_1^2, \quad \ell_3 = c_3 - c_1c_2 + \frac{1}{3}c_1^3.$$

Put $L = \log(x/C)$ and let W_0 be the positive real branch of Lambert's W function. The leading real inverse is

$$N = \frac{L}{W_0(L)}, \quad N \log N = L. \quad (18.2)$$

For large x this is unambiguous. If $D_N = 1 + \log N$, a Taylor expansion of $g_1(N + \Delta) = L$ gives

$$t_1(x) = N - \frac{c_1}{ND_N} + \mathcal{O}\left(\frac{N^{-3}}{D_N^2}\right). \quad (18.3)$$

For example, the first correction cancels c_1/N , and its remaining residual is $\mathcal{O}(N^{-3}/D_N)$; division by the derivative, which is comparable to D_N , proves the displayed error.

For any fixed M , the coefficients $\ell_1, \dots, \ell_M$ produce t_M by solving a monotone scalar equation. Alternatively, substitute $t = N + \Delta$ into

$$(N + \Delta) \log(N + \Delta) + \sum_{j=1}^M \ell_j (N + \Delta)^{-j} = N \log N$$

and determine successive coefficients of Δ in powers of N^{-1} , whose coefficients are rational functions of D_N . The nonzero leading derivative D_N makes this a finite recursive operation at every fixed order. Thus the coefficient algorithm also supplies arbitrary fixed-order continuous inverse expansions.

18.3 Proof of the rounded bracket

For fixed M , equation (18.1) supplies constants D_M, n_M with

$$|\log(a_n/C) - g_M(n)| \leq D_M n^{-M-1} \quad (n \geq n_M).$$

Moreover

$$g'_M(t) = 1 + \log t + \mathcal{O}_M(t^{-2}) > 0$$

for large t . There is therefore a unique sufficiently large solution $t_M(x)$. In a fixed relative neighborhood of t_M , the mean value theorem turns the logarithmic error into a displacement bounded by a constant times $t_M^{-M-1}/(1 + \log t_M)$. Increase the constant A_M so that every sufficiently large integer below $t_M - \eta_M$ has $a_n < x$, while every sufficiently large integer at or above $t_M + \eta_M$ has $a_n \geq x$. Outside that relative neighborhood, monotonicity of g_M gives a much larger gap than the error, so the same conclusions hold. Finally, x eventually exceeds every term in the finite initial segment.

The least permissible integer is consequently bounded by the two ceilings in (11.5). This argument includes $x = a_n$. Since $2\eta_M \rightarrow 0$, the two ceilings eventually differ by at most one. A single ceiling formula is valid only if they agree; otherwise exact comparison with the adjacent sequence values resolves the ambiguity.

Combining (18.3) with the $M = 1$ bracket gives the continuous threshold approximation

$$N - \frac{c_1}{N(1 + \log N)} + \mathcal{O}\left(\frac{N^{-2}}{\log N}\right).$$

A shrinking real error here is not an additive shrinking error for the integer $\nu(x)$ without the rounding qualification. Nor does this report furnish numerical values of A_M, D_M, n_M sufficient to certify a finite-input inverse algorithm. Those are separate issues from the proved asymptotic bracket.

19 Checks and reproducibility

The proof uses exact identities, spectral inequalities, and uniform estimates. The computations below check implementation and algebra; they do not replace any of these arguments.

19.1 Independent exact counting routes

The transfer check constructs (12.3) directly from the substring conditions for every binary word at each bounded n . It constructs E_0, E_1 independently, tests their product, builds the component incidence graph, and compares the transfer, Gram, and tree traces. All arithmetic is integral. The actual-board check instead uses ordinary row masks on $2n$ columns with circular horizontal adjacency. Compatible successive rows avoid vertical and diagonal attacks, and a dynamic program tracks the number of kings through $2n$ rows. Its coefficient of king count n^2 is compared with a_n . It does not use the threshold or caterpillar encoding.

Fresh exact checks reached all transfer, factorization, Gram, and tree identities through $n = 10$ (all 2,046 nonempty binary words in this range), and the independent actual-board dynamic program through $n = 7$. The smaller defaults in the distributed companion are deliberately bounded for convenient replay. Extended bounds are selectable rather than silently included in a potentially long default build. In particular, the first ten positive counts reproduced were

n	a_n	n	a_n
1	4	6	1 118 398
2	32	7	20 984 924
3	344	8	437 500 380
4	4 460	9	10 105 541 204
5	66 532	10	257 860 425 672

These counts agree with the published table [5]. A small factual subset of that table is included with provenance; the entire third-party implementation is not.

19.2 Symbolic certificates and diagnostic residuals

An independent exact rational-function derivation checked (16.5)–(16.7) against (16.3) and the composition moment rules. The generic resolvent and inversion procedure provides a second organization of the calculation. The companion distinguishes exact coefficient identities from optional computer algebra and floating-point evaluations.

For numerical orientation define

$$R_2(n) = n^2 \left(\frac{a_n}{Cn^n} - 1 - \frac{c_1}{n} \right), \quad R_3(n) = n(R_2(n) - c_2).$$

The theorem implies $R_2(n) \rightarrow c_2$ and $R_3(n) \rightarrow c_3$. Using the published value at $n = 800$ gives

$$R_2(800) = 538.8698722011076217595565\dots,$$

$$R_3(800) = -29969.50647515447246172011\dots$$

The discrepancy from the limiting coefficients is expected at this scale. These residuals are noncertified numerical diagnostics and are not a proof by fitting.

[write] **6 October 2026.** *Where the large terms come from.* The package’s fixture (`data/173-square-companion-fixtures-a137432_selected.json`) gives as its `source_url` the OEIS b-file <https://oeis.org/A137432/b137432.txt>, with a SHA-256 and the date 3 October 2026, but holds a_{400} and a_{800} ; the OEIS b-file covers $n = 0, \dots, 384$. This Part’s bibliography [5] and the package’s `173-square-sources-REFERENCES.md` attribute the large terms to Matsuo’s table through $n = 800$ (the entry also links Matsuo’s file “a(1)..a(800)”).

So the field names the wrong file for the two largest terms, unless the b-file was longer on 3 October; which file the SHA-256 belongs to was not determined. The write recomputed $R_2(n)$, $R_3(n)$ from the fixture at $n = 100, 200, 400, 800$ (3000-digit arithmetic): all digits printed above are reproduced, and $R_2 = 381.7, 455.3, 507.0, 538.9$ approach $c_2 + c_3/n = 249.2, 412.8, 494.6, 535.4$, which is consistent with the terms but does not verify them.

19.3 Package contract

The accompanying ZIP contains this editable L^AT_EX source, the PDF, original verification programs, a minimal attributed term fixture, exact coefficient checks, and build instructions. It includes no full third-party book or paper and no downloaded Matsuo source code. The build operates in a newly created output directory and refuses to overwrite an existing target. Hash manifests identify the distributed bytes. Default verification is required to pass with both ordinary Python and optimized Python, so correctness checks do not disappear under `-O`. Optional symbolic and high-precision calculations have explicitly stated dependencies. The final release is checked after extraction into a fresh directory; its replay results are recorded in the package.

20 Source comparison and limits of the audit

The following distinctions are material to the report’s claims.

- **A137432.** The inspected OEIS entry defines the cylindrical maximum-density problem and records both the ratio conjecture and Kotěšovec’s leading equivalent. We retain this credit. No correction expansion or proof is displayed in the inspected entry [2].
- **Matsuo’s exact algorithm.** The linked primary README supplies the binary-word and cyclic-height encoding and an algorithm using $\mathcal{O}(n^4)$ multiprecision operations. We retain this separate credit. The inspected README and source do not discuss the correction expansion [4].
- **The 2013 chess-piece book.** Pages 209–210 discuss this cylinder, bounds, and limit questions. Its lower bound $2(n+1)^n \leq a_n$ holds for $n \geq 1$; the printed strict upper bound $a_n < 2^n(n+1)^n$ needs the restriction $n \geq 2$, since $a_1 = 4$ is equality. The later OEIS constant was posted after that edition. The book’s separate ratio inequality is not used here [1].
- **Planar board results.** Larsen’s *The Problem of Kings* concerns the ordinary planar board, as does the 2025 work identified by DOI 10.2478/rmm-2025-0003. They do not directly establish the cylindrical count treated here [9, 10].
- **Cylindrical king permutations.** Bagno, Eisenberg, Reches, and Sigron count permutations, hence configurations with one king per row and column. This differs from n^2 kings on $4n^2$ squares [11].

The source comparison is bounded to the inspected materials. An earlier overlap search was not independently reproduced and is not used as evidence for an exhaustive priority claim. The exact-count, symbolic-coefficient, and residual checks supporting the present report were independently regenerated.

No exhaustive literature theorem, external referee endorsement, formal verification, submission, repository change, or OEIS edit is claimed. The report presents a self-contained argument and reproducible bounded checks, ready for further independent scrutiny.

[write] **6 October 2026.** The write did not read the book (front matter, “Kotěšovec’s book”); the source notes of this Part say the pages were “rendered and independently inspected” (173-square-sources-REFERENCES.md, item 4). The arithmetic of the third bullet is right, and the remark below proves both printed bounds, with the strict upper bound exactly for $n \geq 2$, and their rectangular forms.

Remark 20.1 ([write], 6 October 2026: two-sided bounds). For all integers $h, w \geq 1$, with $A(h, w)$ the cylinder count of Part I (so that $a_n = A(n, n)$),

$$2(h+1)^w \leq A(h, w) \leq 2^h(h+1)^w.$$

Equality holds in the upper bound if and only if $h = 1$ or $w = 1$, and in the lower bound if and only if $h = 1$. In particular $2(n+1)^n \leq a_n$ for every $n \geq 1$, and $a_n < 2^n(n+1)^n$ for every $n \geq 2$, while $a_1 = 4 = 2^1(1+1)^1$.

Proof. Fix $b \in \{0, 1\}^h$ and $w \geq 1$. By (13.1) (Part I, (3.2)), $\text{tr}(M_b^w) = \text{tr}((C_b C_b^\top)^w) = \sum_i x_i^w$, where $x_1, \dots, x_r \geq 0$ are the eigenvalues of the positive semidefinite matrix $C_b C_b^\top$. Their sum is $\text{tr}(C_b C_b^\top)$, the number of entries equal to one of the zero-one matrix C_b , which is $h+1$: the edges of the tree T_b correspond to the $h+1$ path vertices (Section 13; Part I, Appendix A). For nonnegative reals and $w \geq 1$, $\sum_i x_i^w \leq (\sum_i x_i)^w$, and for $w \geq 2$ equality holds if and only if at most one x_i is nonzero, because the expansion of $(\sum_i x_i)^w$ contains, besides $\sum_i x_i^w$, the nonnegative terms $w x_i^{w-1} x_j$ for $i \neq j$. Hence $\text{tr}(M_b^w) \leq (h+1)^w$. For $w = 1$ equality holds for every b . For $w \geq 2$ it holds if and only if C_b has rank one ($C_b \neq 0$), which by Part I, Appendix A (a connected zero-one rank-one incidence graph is complete bipartite, and a complete bipartite tree is a star), equivalently by Part III, Lemma 27.1, happens exactly when b is constant. Summing over the 2^h words gives the upper bound, with equality if and only if $w = 1$ or every word is constant, that is $h = 1$. For the lower bound, the two constant words contribute $(h+1)^w$ each, and every other word contributes $\text{tr}(M_b^w) \geq (\max_i x_i)^w > 0$; nonconstant words exist exactly when $h \geq 2$. Taking $h = w = n$ gives the statements about a_n . $\square$

21 Further questions

Large order and optimal truncation. The proof treats every fixed expansion order, but it does not control c_j as $j \rightarrow \infty$. Determining their large-order growth, the possible divergence of the series, and an optimal truncation rule would quantify the best accuracy obtainable from the correction expansion. The sizes of c_1, c_2, c_3 alone do not settle these questions.

Effective inverse constants. The rounded bracket in Section 18 has asymptotic existence constants. Explicit values of the remainder constants and onset would turn it into a certified finite-input threshold procedure, with exact adjacent-term comparison only when rounding remains ambiguous. The proof’s spectral and geometric tails provide a possible starting point, but that effective calculation is not supplied here.

Rectangular cylinders and a possible change of regime. For a labeled $2h \times 2w$ horizontal cylinder the maximum is hw kings. The same block geometry gives the exact count

$$a_{h,w} = \sum_{b \in \{0,1\}^h} \text{tr}(M_b^w),$$

where M_b has size $h + 1$ and is defined by the same substring rule. This exact identity follows from the same encoding; the square asymptotic theorem does not automatically extend to all aspect ratios.

Formally, if $w/h \rightarrow \lambda > 0$, a run of length $h - k$ has a leading normalized Perron weight approximately $e^{\lambda(1-k)}$. Summation over its two finite boundaries would then require $2e^{-\lambda} < 1$, or $\lambda > \log 2$. The loss of summability at $\lambda = \log 2$ suggests an aspect-ratio change of asymptotic regime. Its nature, the critical window, and the behavior for $\lambda \leq \log 2$ are prospective questions here, not proved extensions of Theorem 11.1.

[write] 6 October 2026. *Answered for compact aspect ratios by Part I* (Report 174, written later the same day, with the same encoding (2.2)). The change of regime at $\lambda = \log 2$ is real: Theorem 1.2 gives the free energy, the subcritical equivalent for $\lambda < \log 2$ and every fixed order for $\lambda > \log 2$; Theorem 1.3 gives the critical window $w = (\log 2)h + t\sqrt{h}$ to constant order; Theorems 8.1–8.3 give the geometry in the three regimes. The summability condition $2e^{-\lambda} < 1$ sketched above is Part I's (6.2). Still open: $\lambda_h \rightarrow 0$, growing critical windows, and subcritical corrections beyond $\mathcal{O}(h^{-1})$ (Section 31, items 1, 2, 4). The other two questions of this section are items 9 and 3 there.

Part III

Planar maximum density kings: factor degree and pole multiplicity with fixed height asymptotics and inverses

Bundle Report 175, archive `Planar_Kings_Pole_Bounds_and_Inverses_Source.zip`, delivered as `report.tex` (13 pages). Delivered title block “Planar Maximum Density Kings / Factor degree and pole multiplicity with fixed height asymptotics and inverses / Research report 175 / 3 October 2026”. Printed as delivered, with `[write]` notes; Report 175’s Section k is Section $k + 21$ here, and its statements move with their sections; its equations, numbered consecutively through the manuscript, keep their delivered numbers, as does its Table 1. Section labels and Remark 25.2 are added in the write.

Abstract of this Part (as delivered). Let $P_h(w)$ count placements of hw indistinguishable nonattacking kings on an ordinary $2h \times 2w$ board, and let $F_h(z) = \sum_{w \geq 1} P_h(w)z^w$. Every irreducible factor of degree d and exponent e in its reduced rational denominator satisfies

$$d \leq \lfloor h/2 \rfloor + 1, \quad e \leq h - 2d + 3.$$

The proof holds entrywise for the full transfer resolvent. It combines the classical Boolean upper triangular transfer with a smaller positive semidefinite Gram matrix and a count of singular factors along strict Boolean chains. For even h , every occurring factor of degree $h/2 + 1$ is simple, proving the assertion recorded as Conjecture 4 on page 83 of Kotěšovec’s 2013 book. No historical first-proof claim is made.

For each fixed height, partial fractions give an eventual exact finite exponential-polynomial formula whose polynomial degrees satisfy the same degree-sensitive bound. We rederive Wilf’s known positive linear dominant term and obtain its Lambert W inverse with a rigorous two-ceiling threshold bracket. The real uncertainty is exponentially small; the integer bracket eventually has width zero or one. Exact bounded checks and an offline reproducible certificate package accompany the proof.

Reading conventions of Part III (`[write]`)

$P_h(w), F_h$	The ordinary $2h \times 2w$ board, no identification; $F_h(z) = \sum_{w \geq 1} P_h(w)z^w$ (A061594 has offset 0 and adds 1).
$a_{i,j}, b_{i,j}$	The row and the column offset of the king in cell (i, j) : Parts I and II’s x_{ij} and y_{ij} .
q, n, N	$q = b $ the Hamming weight (not $e^{-\lambda}$); $n = h + 1$ (not a board size); $N = (h + 1)2^h$ the number of states.
d, e	d the degree of an irreducible factor or of ρ ; e its exponent (not Euler’s number).
C, G_b, T, D, K	$C = X_0^\top X_1$ is Part I’s C_b ; G_b the smaller Gram matrix; T the full transfer, D its block diagonal, K its strictly upper part.

22 Scope and main conclusions

22.1 The counting convention

Throughout, h, w are positive integers. Squares of the ordinary rectangular board are distinguished, kings are indistinguishable, and a placement is a subset of squares. There is no periodic boundary

identification and no quotient by rotations, reflections, or translations. Nonattacking means that the Chebyshev distance between any two occupied squares exceeds one. Partitioning the board into hw disjoint 2×2 cells proves that at most hw kings fit; this maximum is attained by taking the even-coordinate square in every cell.

We study exactly these maximum placements, with h fixed and w varying:

$$F_h(z) = \sum_{w \geq 1} P_h(w) z^w. \quad (1)$$

All denominator statements refer to the reduced scalar rational function over $\mathbb{Q}$, with denominator normalized to have constant coefficient one. They do not refer to the full characteristic polynomial of a transfer matrix. The distinction matters because scalar boundary cancellation actually occurs.

Theorem 22.1. *For every $h \geq 1$, F_h is rational over $\mathbb{Q}$. If an irreducible polynomial of degree d occurs in its reduced denominator with exponent e , then*

$$d \leq \lfloor h/2 \rfloor + 1, \quad 1 \leq e \leq h - 2d + 3. \quad (2)$$

Every finite pole is positive real. The same upper bounds hold for every reduced entry-denominator of the transfer resolvent constructed in Section 23, and for every fixed rational scalar boundary pairing of that resolvent.

Corollary 22.2. *If $h = 2r$, every occurring irreducible denominator factor of degree $r + 1$ has exponent one. If $h = 2r + 1$, every occurring factor of degree $r + 1$ has exponent at most two.*

The even-height conclusion proves the assertion recorded as Conjecture 4 in Kotěšovec [1, p. 83]. The degree bound establishes only the upper-bound direction of the maximum-degree assertion recorded there as Conjecture 2. It does not prove that a factor of degree $\lfloor h/2 \rfloor + 1$ survives in the planar denominator at every height. Nor does it settle the general low-degree multiplicity assertion on that page.

22.2 Attribution and historical limits

The subset-and-height encoding is present in Knuth’s February 1994 notes [7]. Wilf [8, Lemmas 2–3] gives the binary-word block upper triangular transfer and the scalar sum-of-entries generating function. Brown’s 2021 preprint and 2025 published paper [10] give a subset-and-height recursion whose Corollary 1 explicitly covers rectangles. These constructions are prior work. The component-incidence Gram method is also reused from companion cylindrical analysis; no first-priority claim for that machinery is made.

The mathematical contribution developed here is the degree-sensitive all-pole estimate from the smaller Gram dimension and strict Boolean ranks. A bounded prior-work search did not recover Alekseyev’s full 2011 factor-generation derivation, which is credited in Kotěšovec’s discussion. Thus this report proves a published assertion without assuming that the assertion remains open today, and without claiming the first proof in the literature. Wilf’s leading fixed-height asymptotic, partial fraction extraction, and Lambert W inversion are explicitly treated as prior or standard consequences.

[write] **6 October 2026.** The “companion cylindrical analysis” is Reports 173 and 174, printed here as Parts II and I and written earlier the same day: the factorization $M_b = E_0 E_1$ and the component incidence matrix are Part I’s (3.1) and Part II’s Lemma 13.1, and Part I’s Appendix A already uses the smaller Gram

matrix to bound degrees on the cylinder. What this Part adds for the plane is the count of singular factors along strict Boolean chains (Section 25), needed because the planar transfer is not block diagonal. How the conjectures of Kotěšovec's p. 83 split between the two boards is tabulated in the front matter; the write did not read the book.

23 Geometry and the full transfer

23.1 One king in every cell

Index physical rows and columns from zero. In cell row i and cell column j , write the occupied square as

$$(2i + a_{i,j}, 2j + b_{i,j}), \quad a_{i,j}, b_{i,j} \in \{0, 1\}.$$

In a fixed pair of physical columns, kings in adjacent cell rows conflict exactly when the upper vertical offset is one and the lower offset is zero: their row distance is then one, while their column distance is at most one. Consequently, the vertical offsets are nondecreasing. They have a unique threshold $u \in \{0, \dots, h\}$:

$$a_i(u) = \begin{cases} 0, & i < u, \\ 1, & i \geq u. \end{cases} \quad (3)$$

A column-pair state is therefore (b, u) , where $b \in \{0, 1\}^h$ and $0 \leq u \leq h$. Every such state is valid, so the state count is

$$N = (h + 1)2^h.$$

23.2 The exact adjacency rule

Let (b, u) and (c, v) be states in successive column pairs. The kings in the same cell row force $b_i \leq c_i$ for all i : otherwise their horizontal distance is one and their vertical distance is at most one. Once $b \leq c$ coordinatewise, only attacks between adjacent cell rows remain.

For an upper-left and lower-right pair, the vertical distance can be one exactly when $i \geq u$ and $i + 1 < v$, and the horizontal distance can be one exactly when $b_i = 1$ and $c_{i+1} = 0$. For a lower-left and upper-right pair the conditions are $i + 1 < u$, $i \geq v$, $b_{i+1} = 1$, and $c_i = 0$. It follows that the zero-one transfer is

$$T_{(b,u),(c,v)} = 1 \iff \begin{cases} b \leq c, \text{ and} \\ (b_i, c_{i+1}) \neq (1, 0) \quad (u \leq i \leq v - 2), & u < v, \\ (b_{i+1}, c_i) \neq (1, 0) \quad (v \leq i \leq u - 2), & u > v, \\ \text{no further condition,} & u = v. \end{cases} \quad (4)$$

Empty index intervals impose no condition.

The preceding tests are sufficient as well as necessary. Kings separated by at least two cell rows or two cell columns have a physical coordinate distance at least three. There are no other possible attacks.

Lemma 23.1. *For $w \geq 1$, with $\mathbf{1}$ the all-ones vector of length N ,*

$$P_h(w) = \mathbf{1}^\top T^{w-1} \mathbf{1}, \quad F_h(z) = z \mathbf{1}^\top (I - zT)^{-1} \mathbf{1}. \quad (5)$$

Proof. A valid placement is uniquely a sequence of w valid column-pair states with each successive pair allowed by T . Both boundary states are unrestricted. Summing these paths gives the first identity, and summing the matrix geometric series gives the second. In particular $P_h(1) = (h+1)2^h$. No empty-board convention at $w = 0$ is needed. $\square$

23.3 Diagonal blocks and Boolean rank

Order binary words by nondecreasing Hamming weight $|b|$, breaking ties arbitrarily. The transfer is block upper triangular because $b \leq c$, and distinct words of equal weight cannot have an allowed transition. The diagonal block belonging to b has size $h+1$ and entries

$$(M_b)_{u,v} = \begin{cases} 1, & u < v \text{ and } b_u, \dots, b_{v-1} \text{ is nondecreasing,} \\ 1, & u > v \text{ and } b_v, \dots, b_{u-1} \text{ is nonincreasing,} \\ 1, & u = v, \\ 0, & \text{otherwise.} \end{cases} \quad (6)$$

The strict off-diagonal blocks are essential for the planar problem. Replacing the scalar boundary expression in (5) by a sum of block traces would count a different, periodic model.

24 The smaller Gram matrix

24.1 Monochromatic components

For a fixed word b , consider the path on vertices $0, 1, \dots, h$ whose edge i joins i to $i+1$ and has color b_i . For $k \in \{0, 1\}$ retain just the color- k edges, and let X_k be the vertex-versus-component incidence matrix of that forest. Every row of X_k contains exactly one entry equal to one. Put

$$E_k = X_k X_k^\top, \quad C = X_0^\top X_1. \quad (7)$$

A zero component and a one component intersect in at most one vertex: if they shared two vertices, the unique connecting path would have to consist entirely of both colors. Thus C is a zero-one matrix.

Lemma 24.1. *If $q = |b|$, then*

$$M_b = E_0 E_1 = X_0 C X_1^\top, \quad C \text{ has size } (q+1) \times (h-q+1). \quad (8)$$

There are rectangular integer matrices U, V such that

$$UV = M_b, \quad VU = G_b, \quad (9)$$

where G_b is symmetric positive semidefinite of dimension

$$r_b = \min(q+1, h-q+1). \quad (10)$$

Proof. The (u, v) entry of $E_0 E_1$ counts vertices t in the zero component of u and the one component of v . There is at most one such pivot. When $u < v$, it exists exactly if the intervening edge sequence consists of zeros followed by ones. When $u > v$, reading from v to u gives ones followed by zeros. For $u = v$, the unique pivot is u . These are precisely (6).

A path forest with $h + 1$ vertices and $h - q$ zero edges has $q + 1$ components; the one-edge forest has $h - q + 1$. Finally the two choices

$$\begin{aligned} U = X_0, \quad V = CX_1^\top &\implies VU = CC^\top, \\ U = X_0C, \quad V = X_1^\top &\implies VU = C^\top C \end{aligned}$$

give the claimed factorization. Choose whichever Gram matrix is smaller. $\square$

24.2 Simple finite poles of each diagonal resolvent

For rectangular U, V , the identity

$$(I - zUV)^{-1} = I + zU(I - zVU)^{-1}V \quad (11)$$

follows by direct multiplication wherever the inverses exist, and therefore as a rational matrix identity. Applying it to Lemma 24.1 gives the decisive local information.

Lemma 24.2. *Every finite pole of $(I - zM_b)^{-1}$ is simple at worst and is the reciprocal of a positive eigenvalue ρ of G_b . If $d = [\mathbb{Q}(\rho) : \mathbb{Q}]$, then*

$$d \leq \min(q + 1, h - q + 1), \quad d - 1 \leq q \leq h - d + 1. \quad (12)$$

Proof. The real symmetric matrix G_b is diagonalizable and has nonnegative eigenvalues. Its resolvent consequently has only simple finite poles, even when an eigenvalue is repeated. Nonzero eigenvalues are positive. Identity (11) transfers these properties to the finite poles of the diagonal-block resolvent. The minimal polynomial of ρ divides the characteristic polynomial of the integer matrix G_b , whose degree is r_b . This proves (12). $\square$

One must not substitute a determinant argument for the simple-pole argument: a repeated eigenvalue can give a repeated determinant factor without giving a higher-order resolvent pole. Nor is it necessary to claim that M_b is diagonalizable at zero. A zero-eigenvalue Jordan block contributes a polynomial in z to its resolvent and does not produce a finite pole.

25 Strict Boolean chains control multiplicity

Write

$$T = D + K, \quad D = \text{diag}_b(M_b), \quad R(z) = (I - zD)^{-1}.$$

A nonzero block $K_{b,c}$ requires $b < c$ coordinatewise, hence $|b| < |c|$. A product with more than h such steps vanishes even with arbitrary block diagonal factors inserted. Factoring $I - zT = (I - zD)(I - zRK)$, or multiplying directly, gives the finite expansion

$$(I - zT)^{-1} = R \sum_{k=0}^h (zKR)^k. \quad (13)$$

Each block is a finite sum of products of the form

$$R_{b_0}(z) zK_{b_0,b_1} R_{b_1}(z) \cdots zK_{b_{k-1},b_k} R_{b_k}(z), \quad b_0 < b_1 < \cdots < b_k, \quad (14)$$

where $R_b(z) = (I - zM_b)^{-1}$.

Proof of Theorem 22.1. Rationality follows from (5). Let ζ be a finite pole of an entry of the full resolvent. It is nonzero, since the matrix is analytic at zero. At least one diagonal resolvent in (13) must have a pole at ζ . Thus $\rho = 1/\zeta$ is a positive Gram eigenvalue. Taking a nonzero reciprocal preserves algebraic degree, so put $d = [\mathbb{Q}(\rho) : \mathbb{Q}]$.

By Lemma 24.2, a diagonal factor in (14) can be singular at ζ only at ranks

$$d - 1, d, \dots, h - d + 1.$$

A strict Boolean chain visits each rank at most once. There are $h - 2d + 3$ ranks in this interval, each giving at most a simple pole. Products add pole orders at worst, and finite sums have order at most the largest summand order. Therefore every entry has pole order at most $h - 2d + 3$. The interval is nonempty only if $d \leq \lfloor h/2 \rfloor + 1$.

Fixed rational boundary vectors and the factor z in (5) cannot increase these orders or create new poles. Over $\mathbb{Q}$, irreducible polynomials are separable. In a reduced rational function, the exponent of an irreducible denominator factor equals the pole order at each of its roots. This proves (2) and the entrywise assertion. $\square$

Proof of Corollary 22.2. For $h = 2r$, $d = r + 1$ permits only rank r , so the order is at most one and an occurring factor has exponent exactly one. For $h = 2r + 1$, $d = r + 1$ permits ranks r and $r + 1$, so the order is at most two. $\square$

25.1 Why degree attainment is a separate question

The diagonal block spectrum need not survive the planar scalar pairing. For $h = 3$, some block Gram matrices have eigenvalue one, but the reduced planar denominator is

$$(1 - 4z)^2(1 - 3z)(1 - 4z + 2z^2)^2. \quad (15)$$

Thus the pole at $z = 1$ cancels from F_3 , even though it occurs in individual full-resolvent entries. This directly prevents an argument that simply transfers cylindrical positive trace weights to the planar boundary expression.

For degree one the general estimate gives $e \leq h + 1$. It is deliberately an all-entry bound and is not the conjectured stronger global bound for the scalar low-degree factors. Sharpening it requires information beyond the number of allowable ranks.

Remark 25.1 (A source consistency note). Conjecture 3 on page 83 of [1] prints the bound $\lfloor (h + 1)/2 \rfloor$, while the same page lists exponent two at $h = 1$ and $h = 2$. The exact generating functions below confirm those two exponents. The printed formula is not silently changed to a ceiling here; an intended restriction such as $h \geq 3$ is not supplied by this report. This source issue has no role in the proof of Theorem 22.1.

[write] 6 October 2026. Recorded as this Part states it; the write did not read p. 83. The two exponents are confirmed: F_1 (23) is reduced with the factor $(1 - 2z)^2$ (its numerator is $1 \neq 0$ at $z = \frac{1}{2}$), and is the generating function of $P_1(w) = (w + 1)2^w$; F_2 (24) is reduced with the factor $(1 - 3z)^2$ (numerator $17/9$ at $z = \frac{1}{3}$) and is identical to the generating function of A061593 [12]. The next remark proves the exponent two at every height for the dominant factor.

Remark 25.2 (**[write]**, 6 October 2026: the dominant linear factor). For every $h \geq 1$ the factor $1 - (h + 1)z$ occurs in the reduced denominator of F_h with exponent exactly two. In particular, at $h = 1$ and $h = 2$ some degree-one factor has exponent two, so a bound of value $\lfloor (h + 1)/2 \rfloor = 1$ on

the exponents of degree-one factors would fail there; whether Conjecture 3 bounds exactly these exponents could not be checked without the page.

Proof. Put $n = h + 1$. By Lemma 27.1, the only diagonal blocks with eigenvalue n are those of the two constant words 0^h and 1^h , each simple, and by Lemma 24.2 every finite pole of a diagonal resolvent is simple. In the finite expansion (13) each term (14) is a product along a strict Boolean chain, which visits each of the two constant words at most once; so each term, and hence F_h , has a pole of order at most two at $z = 1/n$ (this is the first sentence of the proof of Theorem 27.2). That proof also shows $A = \lim_{z \rightarrow 1/n} (1 - nz)^2 F_h(z) \geq 1$. So the pole order at $1/n$ is exactly two, and, as in the proof of Theorem 22.1, the exponent of the irreducible factor $1 - nz$ in the reduced denominator equals this pole order. $\square$

26 Exact exponential polynomial formulas at fixed height

All asymptotic statements from this point onward hold with h fixed. Constants may depend on h . No estimate is uniform as h grows with w , and no square-board asymptotic follows from these statements.

Let S_h be the reciprocals of the actual finite poles of the reduced F_h . Every element $\rho \in S_h$ is positive. Write $d_\rho = [\mathbb{Q}(\rho) : \mathbb{Q}]$ and let e_ρ be its pole order.

Theorem 26.1. *There are a polynomial $H_h(z)$ and polynomials $p_\rho(x) \in \mathbb{Q}(\rho)[x]$ such that, for every integer $w \geq 1$ exceeding $\deg H_h$,*

$$P_h(w) = \sum_{\rho \in S_h} p_\rho(w) \rho^w, \quad \deg p_\rho = e_\rho - 1 \leq h - 2d_\rho + 2. \quad (16)$$

If $H_h = 0$, the identity holds for every $w \geq 1$.

Proof. Partial fractions over the real algebraic numbers give

$$F_h(z) = H_h(z) + \sum_{\rho \in S_h} \sum_{j=1}^{e_\rho} \frac{A_{\rho,j}}{(1 - \rho z)^j}. \quad (17)$$

The coefficients belong to $\mathbb{Q}(\rho)$, as can be seen by expanding the rational function locally at $1/\rho$. The top coefficient A_{ρ,e_ρ} is nonzero at an actual pole. Since

$$[z^w](1 - \rho z)^{-j} = \binom{w+j-1}{j-1} \rho^w,$$

we obtain

$$p_\rho(w) = \sum_{j=1}^{e_\rho} A_{\rho,j} \binom{w+j-1}{j-1}. \quad (18)$$

It has degree exactly $e_\rho - 1$, and Theorem 22.1 gives the bound. The polynomial part affects only a finite prefix. $\square$

The restriction to actual scalar poles accounts for all cancellations before the asymptotic formula is formed. The degree-sensitive polynomial bound is the direct new consequence of the pole theorem. The partial fraction method itself is standard.

27 The positive linear dominant term

Wilf [8] proves a positive linear prefactor at the dominant base $h + 1$. We give a direct proof in the present Gram formulation, both to connect the results and to make the inverse analysis self-contained.

Lemma 27.1. *Every nonzero eigenvalue of the full transfer is positive and at most $n = h + 1$. Only the two constant words 0^h and 1^h contribute the eigenvalue n ; each contributes it once. All remaining nonzero eigenvalues are strictly smaller than n .*

Proof. Each path vertex determines its pair of zero and one components, and two such pairs cannot coincide for different vertices. Hence C_b has exactly $h + 1$ entries equal to one and

$$\operatorname{tr} G_b = \|C_b\|_F^2 = h + 1 = n. \quad (19)$$

Its nonnegative eigenvalues are each at most this trace. Equality requires $\operatorname{rank} C_b = 1$.

For a nonconstant word, choose two consecutive edges at a color change. The three incident vertices give, up to row and column permutation, a 2×2 submatrix

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

of C_b . The missing intersection cannot appear elsewhere because the components are intervals of the path. Thus $\operatorname{rank} C_b \geq 2$ and its largest eigenvalue is strictly below n . For either constant word $M_b = J$, where J is the $n \times n$ all-ones matrix; its only nonzero eigenvalue is n , simple. Block triangularity completes the proof. $\square$

Theorem 27.2 (Wilf's leading form). *There are rational constants $\alpha_h \geq 1$ and β_h such that*

$$P_h(w) = (\alpha_h w + \beta_h)(h + 1)^w + R_h(w). \quad (20)$$

If subdominant scalar poles exist and

$$\theta_h = \max(S_h \setminus \{h + 1\}),$$

then $0 < \theta_h < h + 1$ and

$$R_h(w) = O_h(w^h \theta_h^w). \quad (21)$$

Otherwise $R_h(w) = 0$ for all sufficiently large integer w .

Proof. Only the two constant-word diagonal factors can be singular at $z = 1/n$, so the full scalar pole has order at most two. Put $\Pi = J/n$; then

$$\lim_{z \rightarrow 1/n} (1 - nz)(I - zJ)^{-1} = \Pi.$$

The direct off-diagonal block from 0^h to 1^h is also J . The corresponding contribution to F_h in the Boolean-path expansion is

$$z^2 \mathbf{1}^\top (I - zJ)^{-1} J (I - zJ)^{-1} \mathbf{1}.$$

Multiplying by $(1 - nz)^2$ and taking $z \uparrow 1/n$ gives

$$n^{-2} \mathbf{1}^\top \Pi J \Pi \mathbf{1} = 1. \quad (22)$$

All other double-pole path contributions have nonnegative limits: intermediate block resolvents are analytic and entrywise nonnegative at $1/n$, because their spectral radii are smaller than n ; the endpoint residues are Π ; and the connecting blocks are nonnegative. Terms with fewer than two singular factors have zero scaled limit. Consequently

$$A = \lim_{z \rightarrow 1/n} (1 - nz)^2 F_h(z) \geq 1.$$

The dominant partial fractions are $A/(1 - nz)^2 + B/(1 - nz)$, so $\alpha_h = A \geq 1$ and $\beta_h = A + B$. These numbers are rational since $F_h \in \mathbb{Q}(z)$ and $1/n \in \mathbb{Q}$.

All remaining terms in (16) have positive bases below n and polynomial degree at most h , proving (21). The possible polynomial part only affects finitely many widths. $\square$

For any $\theta_h < \eta < h + 1$, the polynomial factor can be absorbed to give $R_h(w) = O_{h,\eta}(\eta^w)$, the familiar exponential-error form. Using the exact subdominant maximum retains the factor w^h .

27.1 Small height examples

At $h = 1$ and $h = 2$ the reduced generating functions are

$$F_1(z) = \frac{4z - 4z^2}{(1 - 2z)^2}, \quad (23)$$

$$F_2(z) = \frac{12z - 29z^2 + 33z^3 - 9z^4}{(1 - 3z)^2(1 - 3z + z^2)}. \quad (24)$$

Thus $P_1(w) = (w + 1)2^w$ for every $w \geq 1$. The factorization of F_3 is given in (15); its numerator in expanded form is

$$32z - 200z^2 + 584z^3 - 1304z^4 + 1624z^5 - 928z^6 + 192z^7.$$

These formulas agree with the classical small-height examples and OEIS entries [12, 13]. For example, the exact dominant constants are

$$(\alpha_1, \beta_1) = (1, 1), \quad (\alpha_2, \beta_2) = (17, -109), \quad (\alpha_3, \beta_3) = (231, -2377).$$

For $h = 4$ they are $\alpha_4 = 7963567/2610$ and $\beta_4 = -12505804889/302760$. The companion certificate records the constants through height six. To extract them rationally, put $n = h + 1$ and $H(z) = (1 - nz)^2 F_h(z)$, analytic at $1/n$. Then

$$\alpha_h = H(1/n), \quad \beta_h = H(1/n) - H'(1/n)/n.$$

These are exact identities and are independently checked by the verifier.

[write] 6 October 2026. Checked by the write against the OEIS (exact rational-function arithmetic): F_2 is the generating function of A061593 [12] identically, whose closed form $(17n - 109)3^n + 2F(2n + 10)$ gives (α_2, β_2) ; A061594 [13] has offset 0 with $a(0) = 1$, and its generating function minus 1 is F_3 , whose explicit formula in the entry has the term $(231n - 2377)4^n$. The extraction formula above reproduces $(17, -109)$ and $(231, -2377)$ from F_2 and F_3 . The values for $h = 4$ to 6 are in `data/175-planar-data-certificates.json` and were not rechecked.

28 Lambert W and integer thresholds

Fix h and abbreviate

$$n = h + 1, \quad b = \log n, \quad \alpha = \alpha_h, \quad \beta = \beta_h.$$

Consider the dominant real function $g(x) = (\alpha x + \beta)n^x$. On its increasing positive branch, its inverse at $y > 0$ is

$$w_0(y) = \frac{1}{b} W_0\left(\frac{by}{\alpha} e^{b\beta/\alpha}\right) - \frac{\beta}{\alpha}, \quad (25)$$

where W_0 is the principal real Lambert function. Indeed, setting $t = b(x + \beta/\alpha)$ transforms $g(x) = y$ into $te^t = (by/\alpha)e^{b\beta/\alpha}$. This is a standard inversion, not a priority claim.

Theorem 28.1. *Suppose subdominant scalar poles exist, and set $r = \theta_h/n \in (0, 1)$. Extend the finite sum in (16) to real x using its positive bases, and denote it by $f_h(x)$. It is eventually strictly increasing. Its eventual real inverse $x_h(y)$ satisfies*

$$x_h(y) = w_0(y) + O_h\left(w_0(y)^{h-1} r^{w_0(y)}\right). \quad (26)$$

For the integer threshold

$$N_h(y) = \min\{w \geq 1 : P_h(w) \geq y\}, \quad (27)$$

there exist $C_h, Y_h > 0$ such that for $y \geq Y_h$, with

$$\varepsilon_h(y) = C_h w_0(y)^{h-1} r^{w_0(y)},$$

one has

$$\lceil w_0(y) - \varepsilon_h(y) \rceil \leq N_h(y) \leq \lceil w_0(y) + \varepsilon_h(y) \rceil. \quad (28)$$

If there are no subdominant scalar poles, then for all sufficiently large y ,

$$N_h(y) = \lceil w_0(y) \rceil. \quad (29)$$

Proof. The finite exponential-polynomial formula and its derivative give, for large real x ,

$$\begin{aligned} f_h(x) &= g(x) + O_h(x^h \theta_h^x), \\ f'_h(x) &= g'(x) + O_h(x^h \theta_h^x), \\ g'(x) &= n^x [\alpha + b(\alpha x + \beta)] \sim \alpha b x n^x. \end{aligned}$$

Hence $f'_h(x) > 0$ eventually. Write $w_0 = w_0(y)$. The dominant differences $g(w_0 \pm 1) - y$ have size comparable to $w_0 n^{w_0}$, with opposite signs, whereas the remainders are smaller by an exponential factor. Thus $x_h(y) \in (w_0 - 1, w_0 + 1)$ for large y . On that interval the derivative is bounded below by a positive constant times $w_0 n^{w_0}$. The mean value theorem applied between w_0 and $x_h(y)$ gives (26).

Every valid width- w arrangement has n distinct extensions obtained by appending the word 1^h with any threshold. The transfer conditions allow all these choices. Therefore

$$P_h(w + 1) \geq n P_h(w) > P_h(w),$$

so the threshold exists and the sequence is strictly increasing. For sufficiently large y , the finitely many exceptional prefix values and the initial nonmonotone portion of the real extension are below the relevant threshold. The exact eventual integer formula then yields

$$N_h(y) = \lceil x_h(y) \rceil.$$

Applying ceilings to the real inverse bounds proves (28). With no subdominant poles, $f_h = g$ eventually, which gives (29). $\square$

28.1 The first explicit subdominant correction

Continue to assume that subdominant scalar poles exist. The exact finite formula also permits one more inversion step. In this subsection, define $R(x) = f_h(x) - g(x)$ using the real exponential-polynomial extension, rather than a bound for an unspecified remainder. Then

$$x_h(y) = w_0(y) - \frac{R(w_0(y))}{g'(w_0(y))} + O_h(w_0(y)^{2h-2}r^{2w_0(y)}). \quad (30)$$

To verify this, put $t = w_0(y)$ and $\delta = x_h(y) - t$. The preceding theorem gives $\delta = O_h(t^{h-1}r^t)$. Taylor's theorem on $[t-1, t+1]$ yields

$$0 = g'(t)\delta + R(t) + O_h(tn^t\delta^2) + O_h(t^h\theta_h^t|\delta|).$$

After division by $g'(t) \asymp_h tn^t$, both error terms are $O_h(t^{2h-2}r^{2t})$, proving (30). Thus the first exponentially small correction is explicitly determined by the subdominant sectors of the exact partial fraction formula. It remains a fixed-height real-inverse estimate and does not remove the rounding issue below.

The real interval before ceilings has exponentially small width $2\varepsilon_h(y)$. Once that width is below one, the integer bracket has width zero or one. Its endpoints agree whenever $w_0(y)$ is farther than $\varepsilon_h(y)$ from every integer. Near an integer boundary, the unconditional replacement $N_h(y) = \lceil w_0(y) \rceil$ is not justified.

The constants in this asymptotic statement are existence constants. A certified answer for a specified finite y needs explicit remainder and derivative bounds, or exact evaluation at the remaining candidate widths. Big- O notation alone is not a finite numerical certificate. For $h = 1$, there is no subdominant term and, for $y \geq 4$,

$$N_1(y) = \left\lceil \frac{W_0(2y \log 2)}{\log 2} - 1 \right\rceil.$$

29 Exact verification and reproducibility

29.1 What the bounded computations establish

The original exact producer checks heights $1 \leq h \leq 6$. It compares all 245 152 ordered state pairs with physical square-coordinate attack tests, checks the component-incidence factorization and smaller Gram determinants, and compares all 36 rectangles $1 \leq h, w \leq 6$ with an independent physical-row mask dynamic program. The reduced denominator degrees are shown in Table 1.

A separately written audit starts with physical square choices and independently reconstructs the transfer. It verifies 7 584 ordered state pairs, all 30 diagonal blocks through height four, and all 16 rectangles through height and width four. Its scalar generating functions come from the full exact characteristic polynomial, without fitted recurrences. A supplementary audit checks all 1 184 resolvent entries through height three, an exact evaluation of the finite Boolean expansion, and the cancellation of $z = 1$ from the scalar $h = 3$ function.

These finite calculations corroborate the proof; they are not a substitute for the all-height argument. The accompanying package distinguishes a short default verification through height four from the explicit extended run through height six.

Table 1: Exact scalar denominator data. Exponent entries are maxima among occurring factors of the indicated degree.

h	States	Denominator degree	$d = 1$	$d = 2$	$d = 3$	$d = 4$
1	4	2	2	—	—	—
2	12	4	2	1	—	—
3	32	7	2	2	—	—
4	80	17	2	2	1	—
5	192	31	2	2	2	—
6	448	75	2	3	2	1

29.2 The all-width recurrence certificate

A fitted prefix alone does not prove a generating function. The certificate used here supplies a finite exact reason that the recurrence holds for all subsequent indices. Put

$$s_m = \mathbf{1}^\top T^m \mathbf{1} \quad (m \geq 0), \quad Q(z) = \sum_{i=0}^D q_i z^i, \quad q_0 = 1,$$

and choose a recorded integer $L \geq D$. The residual at index $L + j$ is

$$\sum_{i=0}^D q_i s_{L+j-i} = \mathbf{1}^\top T^j \left(\sum_{i=0}^D q_i T^{L-i} \right) \mathbf{1}. \quad (31)$$

For an N -state transfer, the certificate checks that (31) vanishes exactly for $j = 0, \dots, N - 1$. Cayley–Hamilton implies that the sequence of these residuals satisfies the characteristic-polynomial recurrence of T . The N consecutive zeros therefore propagate forward to every $j \geq 0$.

The numerator is obtained by convolving Q with the coefficients of $\sum_{m \geq 0} s_m z^m$ at indices below L , and then multiplying by z to obtain F_h . A rational polynomial gcd check certifies that numerator and denominator are coprime. One must retain the distinction between L and D : zero-eigenvalue transients may require $L > D$, and Cayley–Hamilton does not justify propagation backward across them.

29.3 Contents and use of the companion archive

The archive contains this L^AT_EX source, the generated PDF, exact rational generating-function data, an original standard-library verifier, optional exact symbolic regeneration code, and build instructions. Default verification is deliberately bounded. The extended mode explicitly requests heights through six. Checks use exceptions rather than Python `assert`, so optimized execution does not remove them. The package includes tests that deliberately corrupt certificate data and require rejection in both ordinary and optimized Python modes.

The build uses isolated temporary directories, refuses to replace an existing destination, disables shell escape during L^AT_EX compilation, and records file hashes. Replays require local Python and a local L^AT_EX installation; no network access or downloaded implementation is needed. Optional symbolic regeneration uses an already installed SymPy. Third-party books, papers, source code, and research working notes are excluded; primary sources are cited by URL.

[write] 6 October 2026. The code provenance of this package says it was “prepared 2026-10-03 with OpenAI assistance” (producer field of `data/175-planar-data-certificates.json` and `code/175-planar-regenerate.py`; `175-planar-README_CODE.md`). The shipped names and the rerun route are in the README of this report; on a copy of this directory, `python -B code/175-planar-verify.py --data data/175-planar-data-certificates.json` reproduces `data/175-planar-data-default_results.json` (checked at the write).

30 Remaining questions

The theorem leaves several concrete problems.

1. **Maximum-degree attainment.** Does an irreducible factor of degree $\lfloor h/2 \rfloor + 1$ survive the reduced planar boundary pairing for every h ? The present argument proves only the upper bound.
2. **Lower-degree multiplicities.** The rank interval allows $h - 2d + 3$ singular visits. Which visits can actually combine into nonzero scalar highest-order residues, and what stronger bounds hold at small d ?
3. **Cancellation criteria.** Can one characterize algebraically or combinatorially when a Gram eigenvalue disappears from F_h , as eigenvalue one does at $h = 3$?
4. **Height-uniform analysis.** What control on residues, pole separation, and spectral gaps would permit h to grow with w ? Fixed-height exponential errors give no such control by themselves.
5. **Effective inversion.** Can explicit manageable remainder bounds turn the two-ceiling bracket into a practical certificate at a prescribed large y , avoiding full transfer evaluation?
6. **Historical overlap.** Recovering the complete 2011 Alekseyev factor-generation argument remains necessary before any stronger priority statement is defensible.

The proof of the stated all-height inequalities is complete independently of these questions. The unresolved issues concern sharpness, refinement, uniformity, effectivity, and historical attribution.

[write] 6 October 2026. These six questions are items 5, 6, 7, 8, 3 and 10 of Section 31. On item 1: Part I proves cylindrical equality at the heights $h = p - 3$ with the alternating word, but the planar boundary pairing can cancel diagonal-block poles, so that argument does not transfer.

31 Further questions and research

[write] 6 October 2026. This section was added in the write, under Vladimir’s standing rule of 4 October 2026: every claim that the three manuscripts leave unproved, and every question they ask, is stated here as an open question with its source, the argument sketch it came with and what is missing. The delivered question lists (Part I, Appendix C; Part II, Section 21; Part III, Section 30) are printed in their Parts as delivered; the items below re-scope them where another Part answers them. No delivered statement was found to be false.

1. *Vanishing aspect ratio* (Part I, Appendix C, item 1). Theorem 1.1 needs w/h bounded below by a positive constant, and its one-macroscopic-run argument does not cover fixed or slowly growing width. Known at the other end: for fixed h and $w \rightarrow \infty$, $A(h, w) \sim 2(h + 1)^w$ (Appendix A); for

fixed w and $h \rightarrow \infty$ only the small closed forms (B.1), (B.3). *Missing:* the intermediate regimes $\lambda_h \rightarrow 0$.

2. *Growing critical windows and matching* (Part I, item 2). Theorem 1.3 is uniform for bounded $|t_h|$ only; Section 7 shows that G matches the two fixed regimes formally as $t \rightarrow \pm\infty$, and says that this does not extend the remainder. *Missing:* effective uniform remainders for growing $|t_h|$ and the maximal matching ranges.
3. *Effective constants and onsets* (Part I, item 3; Part II, Section 21; Part III, Section 30, item 5). Every remainder constant and onset in the three Parts is existential, so none of the integer brackets (Theorem 10.1, (11.5), (28)) is a certified finite-input procedure. Sketches: the spectral and geometric tails of Part II's proof (Part II); explicit remainder and derivative bounds or exact evaluation at the candidate widths (Part III). *Missing:* explicit constants and onsets.
4. *Subcritical corrections* (Part I, item 4; the residue of Part II's rectangular question). The subcritical equivalent (1.4) has relative error $\mathcal{O}(h^{-1})$; the marked-star sum is explicit, but averaging the spectral perturbation over a long boundary needs a further coefficient analysis, and Section 5 warns that the constants inside (5.3) do not give the first correction. *Missing:* the corrections beyond $\mathcal{O}(h^{-1})$. Part II's question about rectangular cylinders and a change of regime at $\log 2$ (Section 21) is otherwise answered by Part I for every compact range of λ (note there).
5. *The maximal degree at every height* (Part I, item 5; Part III, item 1; Kotěšovec's Conjecture 2). Upper bound $\lfloor h/2 \rfloor + 1$ proved for both boards; equality proved on the cylinder for $h = p - 3$ only, by the alternating word; on the plane the scalar boundary pairing can cancel diagonal-block poles (the pole at $z = 1$ for $h = 3$, (15)), so cylindrical positivity does not transfer. Table 1 shows the top degree $\lfloor h/2 \rfloor + 1$ occurring for $h \leq 6$. *Missing:* attainment at every height, cylinder and plane.
6. *Low-degree multiplicities, and Conjecture 3* (Part III, item 2, and Remark 25.1). Theorem 22.1 allows exponent $h + 1$ for degree one; Table 1 shows 2 for $h \leq 6$, and Remark 25.2 proves exponent exactly two for the factor $1 - (h + 1)z$. *Missing:* which singular visits combine into nonzero highest-order scalar residues; sharper bounds for small d ; the exact statement of Kotěšovec's Conjecture 3 (book not read) and its status.
7. *Cancellation criteria* (Part III, item 3). *Missing:* an algebraic or combinatorial criterion for a Gram eigenvalue to disappear from F_h , as eigenvalue one does at $h = 3$.
8. *Height-uniform planar analysis* (Part III, item 4). Every planar asymptotic statement is for fixed h . *Missing:* control of residues, pole separation and spectral gaps that would allow h to grow with w , the planar analogue of Part I.
9. *Large order of the c_j* (Part II, Section 21). $c_1 \approx -13.6$, $c_2 \approx 576$, $c_3 \approx -32709$ grow quickly; Part II proves every fixed order and nothing about c_j as $j \rightarrow \infty$. *Missing:* the growth of c_j , divergence or convergence of the series, an optimal truncation rule; the same for Part I's $c_j(\lambda)$.
10. *History of the 2011 conjectures* (Part I, item 5; Part III, item 6). The book credits a 2011 factor-generation method to Alekseyev and records his checks through $h = 20$ (Part I); the Zealint discussion was inaccessible. *Missing:* Alekseyev's full argument and any earlier proof of Conjectures 1, 2 (upper half) and 4, before any stronger priority statement.
11. *The book's pages* (added by the write). The write did not read Kotěšovec's book; the conjectures, dates and page numbers in this report are the Parts' readings. *Missing:* a reading of pp. 82–83,

169–196 and 209–210 of the file the Parts used, to confirm the statements of Conjectures 1–4, Part III’s report that Conjecture 3’s printed formula contradicts the listed exponents, and Part II’s report on the strict bound of pp. 209–210.

12. *The data behind the residuals* (Part II, Section 19; the write). The terms $a_{100}, \dots, a_{800}$ are published inputs, not recomputed by any package; the fixture’s `source_url` names the OEIS b-file, which stops at $n = 384$; the residuals $R_2(800)$, $R_3(800)$ are uncertified; Part I’s star-sum diagnostics test $S(h, w)$, not $A(h, w)$. *Missing*: an independent recomputation of the large terms (Matsuo’s algorithm uses $\mathcal{O}(n^4)$ multiprecision operations), the identification of the file whose SHA-256 the fixture records, and certified remainder bounds.

Nothing in the three manuscripts was found to be wrong. Recorded as the sources state them: the strict upper bound of Kotěšovec’s pp. 209–210 fails at $n = 1$ (Part II; proved in Remark 20.1, which also proves it for $n \geq 2$), and the printed formula of Conjecture 3 contradicts the exponents on the same page (Part III; the exponents are confirmed, the page was not read).

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